When Two Thermal Parameters Become One: Structural Non-Identifiability of the Guyer–Krumhansl Relaxation–Nonlocality Pair at the Fourier Resonance

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Abstract

The Guyer–Krumhansl equation models room-temperature heat conduction in heterogeneous media with three coefficients – diffusivity α, flux relaxation time τ_q and nonlocal coefficient κ^2 – routinely obtained by fitting heat pulse experiments. Its solutions coincide with Fourier solutions at $B = \kappa^2/(\alpha\tau_q) = 1$; this cancellation is established, but its consequence for coefficient recoverability has not been examined. This study analyses the rear-face parameter-to-observable map using Fisher information, profile likelihood, Monte Carlo recovery and Bayesian model selection. At $B = 1$ the response is independent of τ_q for any position, time or excitation, with $\partial T/\partial\tau_q = -\alpha\,\partial T/\partial\kappa^2$ exactly; the Fisher matrix is singular and (τ_q, κ^2) is structurally non-identifiable. Away from resonance the degeneracy broadens into a configuration-dependent band, $B \in [0.628, 1.804]$ at a 20% standard-error criterion; at $B = 1$ specifically the Bayesian information criterion selects the Fourier model over Guyer–Krumhansl even for Guyer–Krumhansl data, while α and B remain determined to 1% and 4–10%: the degeneracy affects the (τ_q, κ^2) split alone. Of twelve audited published calibrations, eight lie inside the band and eleven report no uncertainty; two evaluations of the same basalt specimens differ in τ_q by factors of 1.51–3.50 at $R^2 > 0.99$, consistent with the identified structure. The forward operator was validated against three published reference figures without fitting, at normalised errors of 0.384%, 0.382% and 0.483%. Guyer–Krumhansl estimates are best reported in the (α, B) parameterisation, and campaigns targeting τ_q should be designed away from resonance. No experimental validation of the inverse conclusions is claimed.

Keywords: Guyer–Krumhansl equation, structural identifiability, inverse problem, non-Fourier heat conduction, Fisher information, heat pulse experiment

1. Introduction

1.1. Heat conduction beyond Fourier's law

Fourier's law – an instantaneous, local response of flux to temperature gradient – is accurate whenever the characteristic length and time scales of the process are large compared with the mean free paths and scattering times of the energy carriers. Where that separation of scales fails, measured thermal responses depart from the diffusive prediction. Such departures are established at low temperatures in dielectric crystals,

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where second sound and ballistic propagation have been observed, and at the short times and small dimensions encountered in micro and nanoscale devices [1, 2]. Memory-dependent and nonlocal generalisations of Fourier's law of this kind are also incorporated as the thermal constitutive law in coupled thermoelastic and piezothermoelastic wave problems [3, 4, 5, 6], a distinct application in which the coupling of interest is to elastic or piezoelectric deformation rather than to the heat-pulse inverse problem addressed below. They also occur at room temperature in macroscopic specimens whose internal structure is strongly heterogeneous: heat pulse measurements on rocks, metal foams, layered capacitors and composites produce rear-face temperature histories that a single diffusivity cannot reproduce, in samples several millimetres thick under ordinary laboratory conditions [7, 8].

The practical consequence is that a growing class of thermal characterisation problems cannot be closed with one transport coefficient. Once a non-Fourier constitutive law is adopted, the measurement problem changes character: the experiment must determine two or three coefficients rather than one, and the question of whether the available data can determine them separately becomes as important as the question of whether the model can fit the data. That question is the subject of this paper.

1.2. The Guyer–Krumhansl equation and its parameters

Among the generalisations of Fourier's law, the Guyer–Krumhansl (GK) equation occupies a distinctive position. It was derived by Guyer and Krumhansl from the linearised phonon Boltzmann equation for dielectric crystals [9, 10], and it augments the flux equation with two independent coefficients: a relaxation time τ_q associated with momentum-destroying resistive scattering, and a nonlocal coefficient κ^2 associated with momentum-conserving normal scattering, which correlates the heat flux over a finite spatial range. The same structure has since been recovered from Boltzmann transport theory and from nonlinear phonon hydrodynamics, extended to non-constant material parameters, nanoscale boundary conditions and nonlocal vectorial internal-variable formulations, and implemented in dedicated meshless, two-dimensional and device-scale solvers [11, 12, 13, 14]. Within non-equilibrium thermodynamics the same equation arises from internal variable arguments without recourse to kinetic theory [15, 16], which is what licenses its application to heterogeneous media far from the dielectric-crystal setting in which it was first obtained; the closely related ballistic-conductive and hyperbolic extended-variable systems, their operational-method solutions, and their admissibility conditions have been examined in comparable detail [17, 18, 19].

The GK equation is not the only generalisation in use. The Maxwell–Cattaneo–Vernotte (MCV) equation, which retains relaxation and discards nonlocality, remains widely applied [20, 21], as does the dual-phase-lag model [22]. The Nyíri equation [23] retains nonlocality and discards relaxation. These three are exact special cases of the GK equation obtained by constraining its coefficients, which is what makes a like-for-like comparison among them meaningful, and which is exploited in section 8 [24, 1].

1.3. Experimental evidence and the calibration problem

The experimental case for GK behaviour at room temperature rests on heat pulse measurements in which the rear-face temperature history of a millimetre-scale specimen is recorded after a short front-face pulse [7, 8, 25]. Fitting the GK solution to such a record yields estimates of α, τ_q and κ^2 simultaneously, underlying the standard laser flash method, for which analytical solutions and size-effect studies across specimen thickness are available [26, 25, 27].

The literature contains a substantial number of such calibrations, and their reported coefficient pairs vary considerably even among nominally similar specimens: independent evaluations of the same basalt samples have produced relaxation times differing by factors of several [27]. This scatter is usually discussed as experimental variability. One purpose of the present paper is to show that a large part of it is structural: for a wide range of parameter combinations, the data used in these calibrations cannot separate τ_q from κ^2 at all, so the individual values reported are not determined by the measurement even when the fit is excellent. Nanoscale studies – phonon Boltzmann solvers, peridynamic multiscale formulations, physics-informed networks, quasi-ballistic surface-phonon measurements, and ultrafast electron diffuse scattering – extract effective GK-type coefficients by the same logic and inherit whatever identifiability the underlying measurement possesses [28, 29, 30, 31].

1.4. Inverse problems and parameter identifiability

Estimating transport coefficients from temperature histories is an inverse heat conduction problem, a class known to be ill-posed and studied intensively since Beck's foundational work [32]. The modern literature offers many solution strategies – Tikhonov and asymptotical regularisation, physics-informed neural networks, sequential and decoupled frameworks, and analyses of non-uniqueness in the conduction inverse problem itself [33, 34, 35] – addressing *stability*: how to obtain a usable estimate despite noise amplification.

A logically prior question is *identifiability*: whether the parameters are determined by the model-plus-observation structure at all, irrespective of noise and of the estimator used. Structural identifiability asks whether two distinct parameter vectors can produce identical model output; practical identifiability asks whether finite noisy data constrain the parameters to a bounded region. The distinction and its diagnostics are standard in systems biology and dynamical modelling – profile likelihood, Fisher-information conditioning, and Fisher-based optimal experimental design [36, 37, 38, 39, 40, 41, 42, 43, 44, 45] – but have not been applied to the GK coefficient pair. The literature screened for this study contains no structural identifiability analysis of (τ_q, κ^2), no Fisher-information conditioning study of the GK inverse problem, and no quantification of the parameter region in which the pair fails to separate. That is the gap the present work addresses.

1.5. The resonance condition and the present contribution

The GK equation possesses a distinguished parameter combination. When $\kappa^2 = \alpha\tau_q$, its solutions coincide exactly with those of Fourier's law for the same boundary data. This cancellation is known and has been reported repeatedly: it appears in the analytical flash solution of Kovács [26], in the room-temperature experiments of Ván and co-workers [8], in the thermodynamic analyses of Fülöp and co-workers [46, 47], and in numerical studies of generalised heat equations [48, 49]. Fehér and Kovács even restrict their fits to $1 \le B < 3$ in recognition of the difficulty [27]. No priority is claimed for the cancellation itself. It is labelled throughout by the dimensionless resonance number $B = \kappa^2/(\alpha\tau_q)$, so that $B = 1$ denotes the resonance.

What has not been established is the inverse-problem consequence of that cancellation, and this is the contribution of the present paper. The following results are established.

1. An exact sensitivity identity, $\partial T/\partial\tau_q = -\alpha\, \partial T/\partial\kappa^2$, holding pointwise on the resonance ray. The Jacobian of the rear-face observable with respect to (τ_q, κ^2) therefore has rank one and the Fisher matrix is exactly singular, so the pair is *structurally*

non-identifiable at $B = 1$. The identity is proved symbolically and verified numerically.

2. A quantitative map of *practical* non-identifiability away from the exact resonance. Inverting the Fisher standard error gives the band $B \in [0.628, 1.804]$ within which τ_q cannot be recovered to better than 20%, a range spanning nearly a factor of three.
3. Confirmation by three methodologically independent routes: Fisher conditioning, profile likelihood, and a 200-trial Monte Carlo study, which agree throughout the regime where the asymptotic approximation is valid.
4. A model-selection consequence: the Bayesian information criterion selects the Fourier model over the GK model at $B = 1$ even when the data are generated by the GK model itself, recovering the rank deficiency without using any derivative information.
5. An audit of twelve literature-reported calibrations placed in B-space, of which eight fall inside the ill-conditioned band.

The scope of the result is bounded as follows. It concerns the (τ_q, κ^2) split only: the diffusivity α and the resonance number B remain well determined across the whole range, including exactly at $B = 1$, so the data are not uninformative and the GK parameters are not unidentifiable in general. The forward operator is validated against published reference solutions [26], which is external validation of the forward operator and not of the inverse conclusions. No experimental validation is claimed anywhere in this paper.

1.6. Organisation

Section 2 fixes notation and derives the Laplace-domain propagation relation. Section 3 establishes the structural result analytically. Section 4 sets out the numerical methodology, the three independent solvers and the estimation algorithm. Section 5 validates the forward operator against published reference solutions, and section 6 certifies the numerics. Section 7 presents the Fisher, profile-likelihood and Monte Carlo analyses, section 8 the model comparison, and section 9 the audit of published calibrations. Section 11 discusses design implications and section 13 concludes.

2. Guyer–Krumhansl Model and Dimensionless Formulation

This section fixes notation and derives the Laplace-domain propagation relation underlying the identifiability analysis of section 3. The formulation is the one dimensional laser flash problem solved analytically by Kovács [26], which serves as the validation anchor in section 5. No mechanical, electrical or diffusive field is introduced, and no constitutive term beyond those in (1) and (2) appears anywhere in the paper.

2.1. Governing equations

Consider a homogeneous slab of thickness L occupying $0 \le x \le L$, with temperature $T(x,t)$ and heat flux $q(x,t)$. In the absence of internal sources, the balance of internal energy for a rigid conductor at constant density is

$$\rho c \frac{\partial T}{\partial t} + \frac{\partial q}{\partial x} = 0, \tag{1}$$

where ρ is the mass density and c the specific heat, so that ρc is the volumetric heat capacity. Equation (1) is the standard one dimensional energy balance and is not specific to any constitutive law.

The constitutive relation is the Guyer–Krumhansl equation, written in the flux form used by the anchor [26] and originating in the linearised phonon Boltzmann treatment of Guyer and Krumhansl [9, 10],

$$\tau_q \frac{\partial q}{\partial t} + q + \lambda \frac{\partial T}{\partial x} - \kappa^2 \frac{\partial^2 q}{\partial x^2} = 0, \tag{2}$$

in which λ is the thermal conductivity, τ_q is the relaxation time of the heat flux, and κ^2 is the coefficient of the second spatial gradient of the flux. The thermal diffusivity is

$$\alpha = \frac{\lambda}{\rho c}. \tag{3}$$

Three remarks on the physical content of the coefficients are needed, because the identifiability question is a question about these three numbers. The relaxation time τ_q measures the delay with which the flux responds to an imposed temperature gradient; setting $\tau_q = 0$ and $\kappa^2 = 0$ returns Fourier's law. The coefficient κ^2 carries the dimension of a squared length and introduces spatial nonlocality, so that the flux at a point depends on the flux in a neighbourhood rather than on the local gradient alone. In a kinetic reading of the model for dielectric crystals, κ^2 is associated with the phonon mean free path [10]; in the non equilibrium thermodynamic derivation with internal variables the same equation follows from the second law without committing to a carrier mechanism [15], and this second reading is the one appropriate to the room temperature measurements on heterogeneous specimens that motivate the present study [8, 25]. The distinction matters for interpretation but not for the mathematics below, which depends only on $(\alpha, \tau_q, \kappa^2)$ being positive constants.

Notation convention.. Sources differ in how the nonlocal coefficient is written, and the conversions must be stated rather than assumed. The anchor [26] writes the temperature form of the model with a coefficient κ^2 multiplying $\partial_t \partial_{xx} T$, and defines the dimensionless group $\hat{\kappa} = \kappa/L$; the symbol κ^2 in (2) is that same quantity, with dimension m^2. Some experimental papers report instead a length ℓ or a dimensionless ratio, and those are not interchangeable with κ^2 without an explicit conversion. Where published calibrations are used in section 9, the quantity actually tabulated by the original authors is reported as such, and any conversion is flagged. No parameter from one source is silently identified with a similarly named parameter from another.

2.2. Reduction to a single equation

Eliminating T between (1) and (2) gives a single equation for the flux. Differentiating (2) with respect to x and using (1) to replace $\partial_x \partial_x T$ through $\rho c \, \partial_t T = -\partial_x q$ yields, after division by ρc,

$$\tau_q \frac{\partial^2 q}{\partial t^2} + \frac{\partial q}{\partial t} = \alpha \frac{\partial^2 q}{\partial x^2} + \kappa^2 \frac{\partial^3 q}{\partial t \, \partial x^2}. \tag{4}$$

The same elimination performed on T rather than q produces an equation of identical form in T [26]. The two representations are not interchangeable once boundary data are imposed, and this is the first of two points that must be made explicit.

Remark 1 (Flux representation is required for flux boundary data). The laser flash experiment prescribes the heat flux at the irradiated face. The temperature representation of the Guyer–Krumhansl system is convenient only when the boundary data are of Dirichlet type in T; with a prescribed flux the boundary condition is not expressible in T alone without introducing time derivatives of the boundary temperature. The flux representation (4) is therefore used throughout, and T is recovered afterwards by integrating (1) in time [26, 48].

Remark 2 (The initial time derivative is derived, not imposed). Equation (4) is second order in time, so a well posed initial value problem requires both $q(x,0)$ and $\partial_t q(x,0)$. These cannot be prescribed independently: given $q(x,0)$ and $T(x,0)$, the constitutive relation (2) evaluated at $t=0$ determines

$$\tau_q \frac{\partial q}{\partial t}(x,0) = -\,q(x,0) - \lambda \frac{\partial T}{\partial x}(x,0) + \kappa^2 \frac{\partial^2 q}{\partial x^2}(x,0). \tag{5}$$

For the quiescent initial state used here, $T(x,0) = T_0$ and $q(x,0) = 0$ give $\partial_t q(x,0) = 0$. Imposing a value inconsistent with (5) would violate the constitutive law at the initial instant.

2.3. Geometry, boundary and initial conditions

Figure 1 defines the problem: the one-dimensional slab, the pulsed flux at the front face, the adiabatic rear face, the observable rear-face history, and the governing system with its conditions.

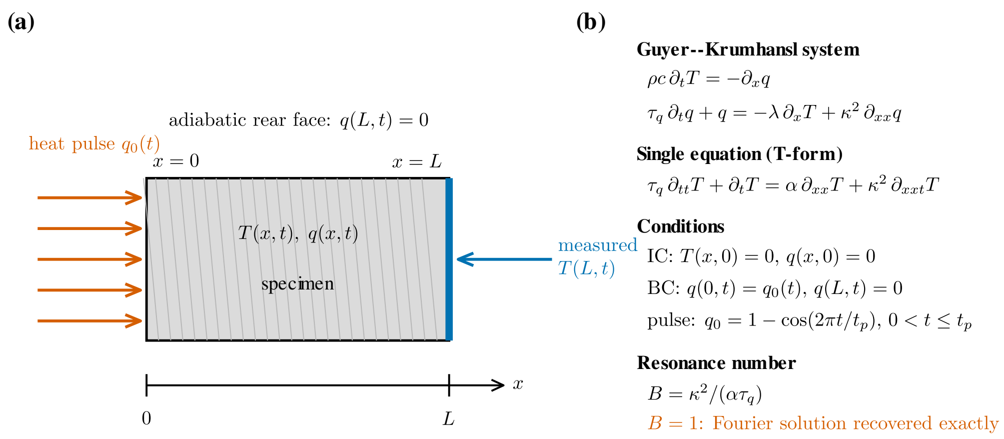

Figure 1: Problem definition. (a) The specimen is a one-dimensional slab of thickness L. The heat pulse $q_0(t)$ is applied at $x=0$, the rear face $x=L$ is adiabatic, $q(L,t)=0$, and the observable is the rear-face history $T(L,t)$. The geometry is genuinely one-dimensional; no lateral or curvilinear dimension enters the model. (b) The Guyer–Krumhansl system, its single-equation T-form, the initial and boundary conditions, the pulse shape, and the resonance number $B = \kappa^2/(\alpha\tau_q)$ at which the Fourier solution is recovered exactly.

The excitation and boundary conditions are those of the validated anchor problem [26] and are reproduced without modification. The irradiated face $x=0$ receives a pulse of finite duration t_p,

$$q(0,t) = \begin{cases} q_{\max}\left[1-\cos\left(\dfrac{2\pi t}{t_p}\right)\right], & 0 < t \le t_p, \\ 0, & t > t_p, \end{cases} \tag{6}$$

where $q_{\max}$ is the amplitude. The rear face is adiabatic for all time,

$$q(L,t) = 0, \tag{7}$$

and the slab starts in equilibrium with its surroundings,

$$T(x,0) = T_0, \qquad q(x,0) = 0, \qquad \frac{\partial q}{\partial t}(x,0) = 0, \tag{8}$$

the last of these following from (5) as required by remark 2 rather than being imposed. Cooling at the boundaries is neglected, as in the anchor [26]. The observable is the rear face temperature history $T(L,t)$, which is the quantity recorded in a flash experiment and the only data used in the inverse problem of section 7.

2.4. Nondimensionalisation

The scaling is that of the anchor [26], reproduced here in full so that every coefficient can be checked. Introduce

$$\hat{t} = \frac{\alpha t}{L^2}, \qquad \hat{x} = \frac{x}{L}, \qquad \hat{T} = \frac{T - T_0}{T_{\text{end}} - T_0}, \qquad \hat{q} = \frac{q}{\bar{q}_0}, \tag{9}$$

in which the time averaged pulse flux and the adiabatic end temperature are

$$\bar{q}_0 = \frac{1}{t_p}\int_0^{t_p} q(0,t)\,\mathrm{d}t = q_{\max}, \qquad T_{\text{end}} = T_0 + \frac{\bar{q}_0 t_p}{\rho c L}. \tag{10}$$

The value $\bar{q}_0 = q_{\max}$ follows because the cosine in (6) integrates to zero over a full period. T_{end} is the temperature the insulated slab reaches once the deposited energy $\bar{q}_0 t_p$ per unit area has spread uniformly, so $\hat{T} \to 1$ as $\hat{t} \to \infty$. The material parameters scale as

$$\hat{\tau}_\Delta = \frac{\alpha t_p}{L^2}, \qquad \hat{\tau}_q = \frac{\alpha \tau_q}{L^2}, \qquad \hat{\kappa}^2 = \frac{\kappa^2}{L^2}, \tag{11}$$

where $\hat{\tau}_\Delta$ is the dimensionless pulse length. Substituting (9) and (11) into (1) and (2) and dropping the hats gives the dimensionless system

$$\tau_\Delta \frac{\partial T}{\partial t} + \frac{\partial q}{\partial x} = 0, \tag{12}$$

$$\tau_q \frac{\partial q}{\partial t} + q + \tau_\Delta \frac{\partial T}{\partial x} - \kappa^2 \frac{\partial^2 q}{\partial x^2} = 0, \tag{13}$$

on $0 \le x \le 1$, with

$$q(0,t) = \begin{cases} 1 - \cos\left(\dfrac{2\pi t}{\tau_\Delta}\right), & 0 < t \le \tau_\Delta, \\ 0, & t > \tau_\Delta, \end{cases} \qquad q(1,t) = 0, \tag{14}$$

and $T(x,0) = q(x,0) = \partial_t q(x,0) = 0$. Eliminating T as in section 2.2 gives the dimensionless counterpart of (4),

$$\tau_q \frac{\partial^2 q}{\partial t^2} + \frac{\partial q}{\partial t} = \frac{\partial^2 q}{\partial x^2} + \kappa^2 \frac{\partial^3 q}{\partial t\,\partial x^2}. \tag{15}$$

Two consequences of (11) are used later and are recorded here. First, the scaling sets the dimensionless diffusivity identically to unity, so α is absorbed into the time variable and cannot be read off from the shape of a dimensionless curve; it is recovered only through the physical time axis. Second, the dimensionless parameters $\hat{\tau}_q$ and $\hat{\kappa}^2$ both depend on L, so a single specimen thickness fixes them jointly. Table 1 collects the symbols, their dimensions and their scaled counterparts.

Table 1: Model parameters, dimensions and scaled counterparts. The dimensionless diffusivity is unity by construction of (9).

Symbol	Meaning	Dimension	Dimensionless form
T	temperature	K	$(T - T_0)/(T_{\text{end}} - T_0)$
q	heat flux	$\mathrm{W\,m^{-2}}$	$q/\bar{q}_0$
x	position	m	x/L
t	time	s	$\alpha t/L^2$
L	slab thickness	m	1
ρc	volumetric heat capacity	$\mathrm{J\,m^{-3}\,K^{-1}}$	not applicable
λ	thermal conductivity	$\mathrm{W\,m^{-1}\,K^{-1}}$	not applicable
α	thermal diffusivity	$\mathrm{m^2\,s^{-1}}$	1
τ_q	flux relaxation time	s	$\alpha\tau_q/L^2$
κ^2	nonlocal coefficient	$\mathrm{m^2}$	κ^2/L^2
t_p	pulse duration	s	$\tau_\Delta = \alpha t_p/L^2$
B	Fourier-resonance number	1	B (invariant)
s	Laplace variable	$\mathrm{s^{-1}}$	conjugate to $\hat{t}$
$m(s)$	propagation factor	$\mathrm{m^{-1}}$	conjugate to $\hat{x}$

2.5. The Fourier-resonance number

The ratio that organises the whole analysis is

$$B = \frac{\kappa^2}{\alpha\tau_q}. \tag{16}$$

It is dimensionless: $[\kappa^2] = \mathrm{m}^2$, $[\alpha] = \mathrm{m^2\,s^{-1}}$ and $[\tau_q] = \mathrm{s}$, so the quotient has dimension $\mathrm{m}^2/(\mathrm{m^2\,s^{-1}} \cdot \mathrm{s}) = 1$. Written in the scaled variables of (11), and recalling that the dimensionless diffusivity is unity,

$$\frac{\hat{\kappa}^2}{1 \cdot \hat{\tau}_q} = \frac{\kappa^2/L^2}{\alpha\tau_q/L^2} = \frac{\kappa^2}{\alpha\tau_q} = B, \tag{17}$$

so B takes the same value in dimensional and dimensionless variables. The slab thickness cancels identically. This scale invariance is used in section 10: within the present one dimensional model, changing the specimen thickness does not move B, and multiple thicknesses of one material share a single value of it.

The quantity B measures the size of the nonlocal contribution relative to the relaxation contribution. It is not introduced here for the first time: the ratio $\kappa^2/(\alpha\tau_q)$ is the quantity Fehér and Kovács restrict to the interval $1 \leq B < 3$ when fitting heat pulse data [27], and the condition $\kappa^2/\tau_q = \alpha$ appears in the anchor [26]. The symbol B is adopted here for compactness.

2.6. Laplace-domain formulation

Let $\hat{f}(x, s) = \int_0^\infty f(x, t)\,\mathrm{e}^{-st}\,\mathrm{d}t$ denote the Laplace transform. Applying it to (12) and (13) with the homogeneous initial conditions (8) gives

$$\tau_\Delta\, s\,\hat{T} + \hat{q}' = 0, \tag{18}$$

$$\tau_q s\,\hat{q} + \hat{q} + \tau_\Delta\,\hat{T}' - \kappa^2\,\hat{q}'' = 0, \tag{19}$$

where a prime denotes $\partial/\partial x$. Differentiating (18) gives $\tau_\Delta s\hat{T}' = -\hat{q}''$, that is $\tau_\Delta \hat{T}' = -\hat{q}''/s$. Substituting into (19),

$$(1+\tau_q s)\,\hat{q} - \frac{\hat{q}''}{s} - \kappa^2\,\hat{q}'' = 0 \qquad \Longleftrightarrow \qquad \hat{q}'' = m^2(s)\,\hat{q}, \tag{20}$$

with the propagation factor

$$m^2(s) = \frac{s\,(1+\tau_q s)}{\alpha + \kappa^2 s}. \tag{21}$$

Equation (21) is written with α retained explicitly, so that it holds in both the dimensional and the dimensionless reading; in the scaling of (11) one sets $\alpha = 1$. The square root $m(s) = \sqrt{m^2(s)}$ is taken with the principal branch, $\operatorname{Re} m > 0$, which is the branch that keeps the solution bounded as the slab thickness grows and is the one required for the inversion contour used in section 4. Since $\tau_q, \kappa^2, \alpha > 0$, the zeros and poles of m^2 lie on the negative real axis at $s = -1/\tau_q$ and $s = -\alpha/\kappa^2$ respectively, so the principal branch is analytic in the right half plane.

Solving (20) with the transformed boundary conditions $\hat{q}(0,s) = \hat{Q}_0(s)$ and $\hat{q}(1,s) = 0$ gives $\hat{q}(x,s) = \hat{Q}_0(s)\,\sinh[m(1-x)]\,/\sinh m$, and recovering the temperature from (18) through $\hat{T} = -\hat{q}'/(\tau_\Delta s)$ yields

$$\hat{T}(x,s) = \frac{m\,\hat{Q}_0(s)\,\cosh[m\,(1-x)]}{\tau_\Delta\, s\,\sinh m}, \tag{22}$$

and, at the observed rear face $x = 1$,

$$\hat{T}(1,s) = \frac{m\,\hat{Q}_0(s)}{\tau_\Delta\, s\,\sinh m}. \tag{23}$$

The transform of the pulse (14) is

$$\hat{Q}_0(s) = \frac{\omega^2\,(1-\mathrm{e}^{-s\tau_\Delta})}{s\,(s^2+\omega^2)}, \qquad \omega = \frac{2\pi}{\tau_\Delta}. \tag{24}$$

The factor $1-\mathrm{e}^{-s\tau_\Delta}$ vanishes at $s = \pm\mathrm{i}\omega$, so the apparent poles of (24) there are removable and $\hat{Q}_0$ is analytic on the imaginary axis. This is not merely a formal observation: section 4 shows that a numerical inversion which separates the two factors reintroduces those poles and fails at a predictable time.

Equations (21) to (23) are the objects the rest of the paper analyses. The parameters τ_q and κ^2 enter the observable (23) only through $m(s)$, and $\hat{Q}_0$ depends on neither. This structure is what makes the resonance analysis of section 3 exact and independent of the pulse shape.

2.7. Fourier, MCV and Nyíri limits

Each competing model used in section 8 is an exact special case of (21), obtained by constraining the parameters rather than by changing the solver. Taking limits directly in (21):

$$\text{Fourier } (\tau_q \to 0,\ \kappa^2 \to 0): \qquad m^2(s) \to \frac{s}{\alpha}, \tag{25}$$

$$\text{MCV } (\kappa^2 \to 0,\ \tau_q > 0): \qquad m^2(s) \to \frac{s\,(1+\tau_q s)}{\alpha}, \tag{26}$$

$$\text{Nyíri } (\tau_q \to 0,\ \kappa^2 > 0): \qquad m^2(s) \to \frac{s}{\alpha+\kappa^2 s}. \tag{27}$$

Equation (25) is the transform of the diffusion equation, for which the response is governed by α alone. Equation (26) is the Maxwell–Cattaneo–Vernotte limit: the extra factor $(1+\tau_q s)$ is quadratic in s, corresponding to the hyperbolic equation $\tau_q \partial_{tt} T + \partial_t T = \alpha \partial_{xx} T$ and to a finite propagation speed $\sqrt{\alpha/\tau_q}$, so this limit produces a travelling front rather than a diffusive rise. Equation (27) retains the nonlocal term alone [23]; it is included in the model comparison because it isolates the mechanism that MCV omits, and it is retained for that reason only. Each limit is dimensionally consistent, since $[m^2] = \mathrm{m}^{-2}$ in every case. The Maxwell–Cattaneo–Vernotte and Nyíri forms therefore test the two mechanisms of the Guyer–Krumhansl model separately, which is what makes the comparison in section 8 a like for like one. Models requiring a variable absent from (21) are excluded from that comparison, as explained there.

2.8. The resonance condition

Setting $B = 1$ in (16) gives

$$B = 1 \quad \Longleftrightarrow \quad \kappa^2 = \alpha\tau_q, \tag{28}$$

which is the Fourier resonance condition. It is established in the Guyer–Krumhansl literature: it is described as a hierarchical resonance at which the solutions of the coupled system may coincide with those of the single Fourier equation [46], it is named as the Fourier resonance and stated in the form $\kappa^2/\tau_q = \alpha$ in the anchor [26], and the same coincidence is described as the Fourier solutions being covered by the Guyer–Krumhansl solutions [47]. It is not claimed as a result of the present work. Its consequences for the recoverability of τ_q and κ^2 from a measured temperature history are derived in section 3, and no part of that derivation is anticipated here.

2.9. Parameterisation

The forward model is defined by the parameter vector

$$\boldsymbol{\theta} = (\alpha,\ \tau_q,\ \kappa^2), \tag{29}$$

which is the set an evaluation procedure is normally asked to return. For the analysis it is convenient to use the equivalent set

$$(\alpha,\ \tau_q,\ \kappa^2) \ \longmapsto\ (\alpha,\ \tau_q,\ B), \qquad \kappa^2 = \alpha\,\tau_q\,B, \tag{30}$$

the inverse transformation being immediate from (16). The map is a smooth bijection on the positive octant $\alpha, \tau_q, \kappa^2 > 0$, with Jacobian determinant $\partial\kappa^2/\partial B = \alpha\tau_q \neq 0$, so no information is created or destroyed by the change of variables. Its usefulness is that the resonance (28) becomes the single coordinate hyperplane $B = 1$ rather than a curved surface in (τ_q, κ^2) space, which makes the degeneracy analysed in section 3 explicit in the coordinates themselves. A change of variables of this kind is a standard device and is not presented as a contribution; what section 7 reports is the quantitative behaviour of the estimation problem in these coordinates.

2.10. Correspondence with the validation anchor

Because section 5 reproduces published figures, the correspondence between the present notation and that of the anchor must be exact. Table B.1 states it. The dimensionless system (12)–(15) is the anchor's system, the scaling (9)–(11) is the anchor's scaling, and the boundary and initial conditions (14) are the anchor's conditions. The anchor solves the problem by eigenfunction expansion; the present work solves the same problem in the Laplace domain and by finite differences, as described in section 4, which is why (21) does not appear in the anchor even though the problem it describes is the same. The governing equation (4) is linear and constant-coefficient, so an eigenfunction solution exists for any fixed parameter set and is exactly what the anchor computes; the choice made here is not dictated by any difficulty of the forward problem itself. It is dictated by the inverse problem: the identifiability analysis of sections 3 and 7 and the model comparison of section 8 require the solution and its analytic derivatives with respect to $(\alpha, \tau_q, \kappa^2)$ at a large number of parameter combinations, whereas an eigenfunction series has eigenvalues defined only implicitly, by a transcendental characteristic equation in those same parameters, and differentiating the series term by term with respect to them is correspondingly indirect. The Laplace-domain transfer function (21) is a closed-form rational function of $(\alpha, \tau_q, \kappa^2)$ and differentiates in closed form at the same cost as the solution itself (section 4.4), which is what the repeated evaluation underlying sections 7 and 8 requires. Equation numbers are not cross referenced between the two documents, because the version of record and the open preprint of the anchor number their equations differently; statements are attributed to the source as a whole.

3. Analytical Structural Identifiability

The consequence of the resonance condition (28) for the inverse problem is derived below. The result is exact and is obtained analytically from the formulation of section 2; the numerical work reported later confirms it but plays no part in establishing it.

3.1. Definition of structural identifiability

Definition 1 (Structural identifiability)**.** Let $\mathcal{M}(\boldsymbol{\theta})$ denote the map from a parameter vector $\boldsymbol{\theta}$ to the noise free observable, here the rear face temperature history. A component or combination of $\boldsymbol{\theta}$ is structurally identifiable if $\mathcal{M}(\boldsymbol{\theta}_1) = \mathcal{M}(\boldsymbol{\theta}_2)$ for all times implies that the component agrees in $\boldsymbol{\theta}_1$ and $\boldsymbol{\theta}_2$. Structural non-identifiability is a property of the model and the observation operator alone: it cannot be removed by reducing noise or by sampling more densely.

Practical identifiability, which concerns estimation to a stated precision from finite noisy data, is a separate matter and is treated in section 7. The two are kept distinct throughout.

3.2. Cancellation on the resonance ray

Proposition 1 (Fourier-equivalence of the propagation factor)**.** *For $\alpha, \tau_q > 0$, the condition $B = 1$ is equivalent to $\kappa^2 = \alpha\tau_q$, and under it the propagation factor (21) reduces identically to that of the Fourier equation:*

$$m^2(s)\Big|_{B=1} = \frac{s\,(1+\tau_q s)}{\alpha + \alpha\tau_q s} = \frac{s\,(1+\tau_q s)}{\alpha\,(1+\tau_q s)} = \frac{s}{\alpha}, \qquad \text{for all } s. \tag{31}$$

Proof. Substituting $\kappa^2 = \alpha\tau_q$ into the denominator of (21) gives $\alpha + \kappa^2 s = \alpha(1+\tau_q s)$. Since $\tau_q > 0$, the factor $(1+\tau_q s)$ is not identically zero and cancels against the identical factor in the numerator, leaving s/α. The cancellation is algebraic and holds for every s in the domain of analyticity, not asymptotically or approximately. □

The single factor $(1+\tau_q s)$ carries the entire dependence of m^2 on the relaxation time. When it cancels, τ_q disappears from the propagation factor, and with it from every quantity built on the propagation factor. Comparison with (25) shows that (31) is exactly the Fourier propagation factor, which is the analytical content of the known resonance [46, 26, 47].

3.3. Independence of the observable from the relaxation time

Proposition 2 (Structural non-identifiability of (τ_q, κ^2))**.** *Consider the Guyer–Krumhansl system of section 2 with $\kappa^2 = \alpha\tau_q$, α fixed. Then:*

1. *the temperature response $T(x,t)$ is independent of τ_q at every position $x \in [0,1]$ and every time t, for any pulse $\hat{Q}_0(s)$;*
2. *consequently the one parameter family $\{(\alpha, \tau_q, \alpha\tau_q) : \tau_q > 0\}$ produces identical observations, so τ_q and κ^2 are not separately structurally identifiable in the sense of definition 1;*
3. *the parameter sensitivities are exactly anti-parallel,*
$$\frac{\partial T}{\partial \tau_q} = -\,\alpha\,\frac{\partial T}{\partial \kappa^2}, \tag{32}$$
so the sensitivity matrix with respect to (τ_q, κ^2) has rank one, the associated Fisher information matrix is singular, and the correlation between the two parameters is -1.

Proof. (i) By (22), the parameters τ_q and κ^2 enter $\hat{T}(x,s)$ only through $m(s)$, since $\hat{Q}_0$ depends on the pulse alone and τ_Δ and s are independent of them. By proposition 1, $m^2|_{B=1} = s/\alpha$, hence $m|_{B=1} = \sqrt{s/\alpha}$ on the principal branch, which contains no τ_q. Therefore
$$\hat{T}(x,s)\Big|_{B=1} = \frac{\hat{Q}_0(s)\,\sqrt{s/\alpha}\,\cosh\!\Big[\sqrt{s/\alpha}\,(1-x)\Big]}{\tau_\Delta\, s\,\sinh\!\sqrt{s/\alpha}}, \tag{33}$$
which is manifestly free of τ_q, so $\partial\hat{T}/\partial\tau_q|_{B=1} \equiv 0$ for all s and all $x \in [0,1]$. The Laplace transform is injective on continuous functions, so the same holds for $T(x,t)$ at every t.

(ii) Immediate from (i): varying τ_q while holding $\kappa^2 = \alpha\tau_q$ traces the stated one parameter family, along which the observable is constant. The map $\mathcal{M}$ is therefore not injective.

(iii) Write $\hat{T} = G(m)$ with $G(m) = \hat{Q}_0 m\cosh[m(1-x)]/(\tau_\Delta s\sinh m)$. By the chain rule $\partial\hat{T}/\partial p = G'(m)\,\partial m/\partial p$ for $p \in \{\tau_q, \kappa^2\}$, so wherever $G'(m) \neq 0$ the ratio of sensitivities depends only on m:
$$\frac{\partial\hat{T}/\partial\tau_q}{\partial\hat{T}/\partial\kappa^2} = \frac{\partial m/\partial\tau_q}{\partial m/\partial\kappa^2}. \tag{34}$$
Differentiating (21),
$$\frac{\partial m}{\partial \tau_q} = \frac{s^2}{2m\,(\alpha+\kappa^2 s)}, \qquad \frac{\partial m}{\partial \kappa^2} = -\,\frac{m\,s}{2\,(\alpha+\kappa^2 s)}, \tag{35}$$
whence the ratio (34) equals $-s/m^2$ in general. Substituting $m^2 = s/\alpha$ from proposition 1 gives the value $-\alpha$, independent of s and of x. Because the ratio is the same constant at every s, it is the same constant at every t after inversion, which is (32). Stacking the

sensitivities evaluated at the observation times into a matrix $J = [\,\partial T/\partial\tau_q \;\; \partial T/\partial\kappa^2\,]$, the two columns satisfy $J_{:,1} = -\alpha J_{:,2}$, so $\operatorname{rank} J = 1$ and $J\,(1,\;\alpha)^{\top} = \mathbf{0}$. The Fisher information matrix for independent Gaussian errors of variance σ^2 is $F = J^{\top} J/\sigma^2$, which inherits the rank deficiency, so $\det F = 0$ and the implied correlation is -1. □

Three features of proposition 2 deserve emphasis. First, it holds for any pulse: $\hat{Q}_0(s)$ cancels from the ratio (34), so no choice of excitation waveform restores identifiability. Second, it holds at every interior position, not only at the rear face, so moving or adding thermometers within the slab does not help. Third, the null direction $(1,\alpha)$ in the (τ_q,κ^2) plane is precisely the tangent to the ray $\kappa^2 = \alpha\tau_q$, confirming that the flat direction of the likelihood is the resonance ray itself.

3.4. Identifiable parameter combinations

proposition 2 is a statement about the pair (τ_q,κ^2) and must not be read more broadly. The diffusivity α is unaffected: by (33) the response at resonance is exactly a Fourier response with diffusivity α, and a Fourier response determines α from the physical time axis. Likewise the combination B remains meaningful, since the statement $B = 1$ is itself an identified fact about the data. In the parameterisation (30) the degeneracy is confined to a single coordinate:

$$\left.\frac{\partial T}{\partial\tau_q}\right|_{\alpha,\,B} \;=\; \frac{\partial T}{\partial\tau_q} + \alpha B\frac{\partial T}{\partial\kappa^2} \;\overset{B=1}{=}\; \frac{\partial T}{\partial\tau_q} + \alpha\frac{\partial T}{\partial\kappa^2} \;=\; 0, \tag{36}$$

by (32), while the derivatives with respect to α and B are not annihilated. The information the data carry is therefore carried by (α, B), and it is the split of that information into τ_q and κ^2 separately that fails. The correct conclusion is not that Guyer–Krumhansl parameters cannot be identified, but that the identifiable content of a temperature record depends on where the material sits relative to the resonance. Section 7 quantifies this for finite noisy data and shows that α and B remain well determined even at $B = 1$.

3.5. Independent verification

The proof above is analytic. It was checked independently in three ways, each of which confirms rather than establishes it. Symbolic computer algebra applied to (21) and (22) reproduces $m^2|_{B=1} = s/\alpha$, returns identically zero for $\partial\hat{T}/\partial\tau_q$ on the ray at arbitrary x, and simplifies the sensitivity ratio at resonance to exactly $-\alpha$; the general off resonance ratio is returned as $-s/m^2$, in agreement with (34) and (35). Evaluation in 50 digit arithmetic along the ray, with τ_q varied over five orders of magnitude, leaves the computed response invariant to a spread below 10^{-30}, whereas a control at $B = 1.05$ separates by order 10^{-2}, so the collapse is a property of the ray and not of the arithmetic. Finally, an independent finite difference solver that never forms a Laplace transform reproduces the same invariance. Details of the numerical framework are given in section 4 and its certification in section 6. The collapse is exhibited directly in fig. 2: three responses whose relaxation times differ by two orders of magnitude are indistinguishable on the resonance ray, while the same three separate at $B = 1.05$.

Remark 3 (Relation to prior work). The coincidence of Guyer–Krumhansl and Fourier solutions at $\kappa^2 = \alpha\tau_q$ is established [46, 26, 47, 27] and is used here as a starting point, not offered as a finding. What proposition 2 adds is the inverse problem statement: that the coincidence implies an exact rank deficiency of the sensitivity matrix with the explicit null direction $(1,\alpha)$, hence a singular Fisher information matrix and the structural

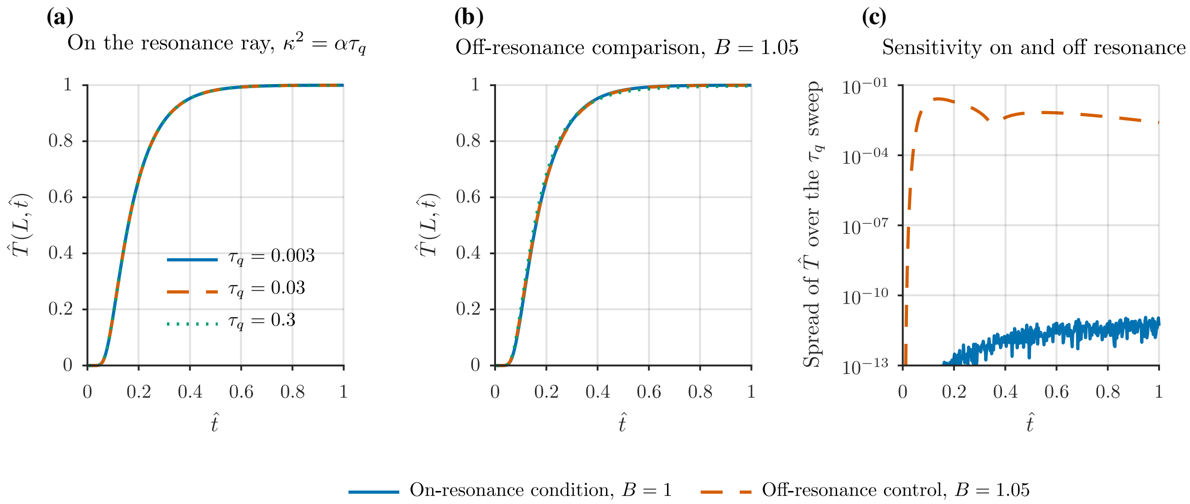

Figure 2: Structural degeneracy of the (τ_q, κ^2) pair in the observable. (a) Rear-face temperature history computed along the resonance ray $\kappa^2 = \alpha\tau_q$, with τ_q varied over a factor of 100; the three solutions are graphically indistinguishable. (b) The same sweep at the off-resonance value $B = 1.05$. (c) Spread of $\hat{T}$ across the τ_q sweep for both conditions on a logarithmic ordinate: 1.156×10^{-11} on the resonance ray against 2.595×10^{-2} at the off-resonance condition, a ratio of 2.24×10^{9}.

non-identifiability of the pair (τ_q, κ^2) from any temperature observation, at any noise level and any sampling density.

4. Numerical Methodology

Three independent solvers of the system of section 2 are used. Their agreement is treated as numerical cross-checking, not as validation; external validation is the separate exercise of section 5. Every numerical setting reported below was fixed by the convergence and stability study of section 6 and is stated with its justification.

4.1. Solver A: Laplace inversion by convolution

The primary solver evaluates (23) numerically. A direct inversion of (23) is not used, for a reason that is a result of this work and is documented in section 6. The transform factorises as

$$\hat{T}(1,s) = \hat{K}(s)\,\hat{Q}_0(s), \qquad \hat{K}(s) = \frac{m(s)}{\tau_\Delta\, s\, \sinh m(s)}, \tag{37}$$

into a material kernel $\hat{K}$ and the pulse transform $\hat{Q}_0$ of (24). As noted after (24), $\hat{Q}_0$ has removable singularities at $s = \pm i\omega$. The kernel $\hat{K}$ has none. Accordingly the kernel alone is inverted numerically and the temperature is recovered by the exact convolution

$$T(1,t) = \int_0^{\min(t,\tau_\Delta)} k(t-u)\, q_0(u)\, \mathrm{d}u, \qquad q_0(u) = 1 - \cos\!\left(\frac{2\pi u}{\tau_\Delta}\right), \tag{38}$$

k being the inverse transform of $\hat{K}$. The integrand is analytic on the short support $[0, \tau_\Delta]$, so a fixed order Gauss–Legendre rule converges rapidly; 32 nodes are used, at which the quadrature is converged to better than 10^{-13} (doubling to 64 nodes changes the result by at most that amount).

The kernel is inverted by the fixed Talbot rule [50, 51], which evaluates $\hat{K}$ at the nodes

$$s_k = r\,\theta_k\,\left(\cot\theta_k + \mathrm{i}\right), \qquad \theta_k = \frac{k\pi}{M}, \qquad r = \frac{2M}{5t}. \tag{39}$$

The number of nodes is $M = 41$. Two constraints fix this choice. The value must be odd: at $\theta = \pi/2$, attainable only when M is even, the node lies on the imaginary axis, and section 6 shows that an even M places a quadrature node exactly on the removable pulse pole at a predictable time. The value must also be moderate, because the rule loses accuracy to round off for large M; the accuracy plateau extends to about $M \approx 81$ and degrades beyond it. Two further implementation details are required to avoid overflow, since $\operatorname{Re} m$ becomes large on the contour: $1/\sinh m$ is evaluated as $2\mathrm{e}^{-m}/(1-\mathrm{e}^{-2m})$, and $\coth m$ as $(1+\mathrm{e}^{-2m})/(1-\mathrm{e}^{-2m})$.

4.2. Solver B: staggered-grid finite differences

The second solver integrates (12) and (13) directly in the time domain, with no Laplace transform, so it shares no code path and no approximation with Solver A. Temperatures are stored at N_x cell centres and fluxes at the $N_x + 1$ cell faces, which places the boundary conditions (14) exactly on grid points and yields second order accurate centred differences for both $\partial_x q$ and $\partial_{xx} q$. The resulting system of ordinary differential equations is stiff, and is integrated with a backward differentiation formula at relative tolerance 10^{-10} and absolute tolerance 10^{-12}. Production runs use $N_x = 400$, for which section 6 reports a spatial discretisation error of order 10^{-6}.

4.3. Solver C: arbitrary-precision reference

The third solver evaluates the same transform in arbitrary precision arithmetic at 50 significant digits, and serves only as a ground truth for certifying the other two at selected probe points. Its inversion degree must be at least 24: at lower degree it returns a converged looking but incorrect value, an error of about 2.5% having been observed at degree 10 near the end of the pulse.

4.4. Analytic sensitivities

All statistical quantities require the derivatives of the predicted temperature with respect to the parameters. Finite differencing the time domain solver is avoided, because differencing at the solver noise floor contaminates the Jacobian and produces a spurious discrepancy between the Fisher and Monte Carlo uncertainty estimates. The sensitivities to τ_q and κ^2, the parameters whose identifiability is at issue, are instead obtained analytically in the Laplace domain. Differentiating (37) through m and using (35),

$$\frac{\partial \hat{K}}{\partial m} = \frac{1}{\tau_\Delta\, s\, \sinh m}\left(1 - m\coth m\right), \qquad \frac{\partial \hat{K}}{\partial p} = \frac{\partial \hat{K}}{\partial m}\,\frac{\partial m}{\partial p}, \qquad p \in \{\tau_q, \kappa^2\}, \tag{40}$$

and these are inverted and convolved by exactly the procedure of (38). Because α rescales the time axis and the pulse length rather than entering the kernel as a parameter, its sensitivity is obtained by a central difference in α alone. The analytic derivatives were verified against central differences of the forward solver, agreeing to a maximum relative deviation of 5.3×10^{-7}.

4.5. Estimation and statistical procedures

Parameters are estimated by Levenberg–Marquardt minimisation [32] of the sum of squared residuals, carried out in logarithmic parameters so that positivity is enforced automatically and the scaling of the three parameters is comparable, with the analytic Jacobian of section 4.4. Synthetic observations are generated by adding independent Gaussian noise of standard deviation $\sigma = \eta\,(T_{\max} - T_{\min})$ to the noise free response, with $\eta = 2\%$ unless stated otherwise.

Four diagnostics are used, and are reported together because each fails in a different way near a degeneracy. The Fisher information matrix for independent Gaussian errors is $F = J^\top J/\sigma^2$, with J the analytic sensitivity matrix in log parameters. Asymptotic standard errors are the square roots of the diagonal of F^{-1}, and the condition number of F measures the conditioning of the estimation problem. This is the standard construction underlying Fisher-based experimental design [44, 52, 41], and the way the matrix is approximated is known to affect the resulting design [43]. The Fisher analysis is local and assumes a quadratic log likelihood, so it is checked against a Monte Carlo study, in which the estimator is applied to repeated noise realisations and the empirical spread of the estimates recorded. Profile likelihood [36, 37], computed by fixing τ_q on a grid and re optimising the remaining parameters, gives a confidence interval valid without the quadratic assumption, using the $\chi^2(1)$ threshold of 3.841 at the 95% level; workflows automating this diagnostic are now available [38, 39]. Model selection uses the Bayesian information criterion [53], $\mathrm{BIC} = n\ln(\mathrm{SSE}/n) + k\ln n$, applied to candidate models that are exact special cases of the same transfer function (21) so that solver, protocol and convergence criteria are identical across models.

Because the estimation problem is poorly conditioned near resonance, the dependence of the result on the optimiser starting point is itself studied: starts are drawn from log normal perturbations of the true parameters at several widths, and the spread of the recovered values is recorded separately from the spread induced by measurement noise.

4.6. Solution algorithm

The forward evaluation, the sensitivity evaluation and the identifiability diagnostics are organised into the single procedure summarised in algorithm 1 (Appendix A). The structure is dictated by two requirements established in section 6: the material kernel must be inverted separately from the pulse transform, because the split form fails at even Talbot node counts, and the sensitivities must be obtained analytically rather than by finite differences, because differencing at the solver noise floor corrupts the Fisher matrix.

Three features of the procedure are dictated by numerical requirements rather than by convenience, and are justified below.

Kernel-only inversion (line 4). The pulse transform $\hat{Q}_0$ has removable singularities on the imaginary axis. When the full product is inverted on a Talbot contour with an even number of nodes, a quadrature node coincides with one of these singularities and the computed temperature diverges by up to ten orders of magnitude at a predictable instant. Inverting the pole-free kernel alone removes the failure mode entirely, and the parity dependence is documented in fig. 5. Talbot's method itself [50] and its modern refinements [51] are well established; the interaction with a pulse transform carrying removable poles is what required care here. Earlier treatments of Laplace inversion for inverse conduction [54] and analyses of inversion stability [55, 56] informed the choice of contour.

Analytic sensitivities (line 11). Because $m^2(s)$ is a rational function of s with polynomial dependence on the parameters, each sensitivity can be inverted on the same contour at the same cost as the solution itself. Finite-difference Jacobians formed at the solver noise floor yield a systematic discrepancy between the Fisher and Monte Carlo uncertainty estimates; with analytic sensitivities the two estimates agree, as reported in fig. 7(d). The influence of the Fisher-matrix approximation on the resulting inference is documented in the estimation literature [43].

Profiling rather than linearised intervals (line 16). Near the resonance the log-likelihood ceases to be quadratic in τ_q, so an asymptotic standard error is no longer a meaningful summary. Profile likelihood is the standard remedy [36, 37] and is what reveals that the admissible interval is unbounded at $B = 1$ rather than merely large.

5. External Validation of the Forward Operator

External validation of the forward operator against an independently published solution is reported here. The anchor is the analytical laser flash solution of the Guyer–Krumhansl equation published by Kovács [26], which solves exactly the problem set out in section 2 by eigenfunction expansion, independently of the methods of section 4.

5.1. Categories of evidence

Two categories are kept strictly separate throughout this paper. Agreement between independent implementations of the same equations is numerical cross-checking, not external validation. Agreement between the present implementation and results published by another author is external validation. The present section reports the second; the first is reported in section 6.

A third distinction is equally important. What is validated here is the *forward operator*: the ability of the present implementation to compute the temperature response of the Guyer–Krumhansl system given the parameters. This validation makes no statement about the inverse problem. The identifiability results of sections 3 and 7 are established analytically and verified by the separate means described there, not by this reproduction. No experimental validation of the inverse conclusions is claimed anywhere in this paper.

5.2. Reproduction protocol

Published figures were extracted from the anchor at native resolution (4000×1908 pixels) and digitised by two independent passes, one taking the intensity weighted centroid of the plotted curve band and one taking the midpoint of its envelope. The disagreement between the passes, combined with the curve half width, defines a digitisation noise floor

$$\sigma_d = \sqrt{\overline{(\Delta_{AB})^2} \; + \; \text{median(half width)}^2}, \tag{41}$$

where Δ_{AB} is the pointwise difference between the two passes. All comparisons are reported against σ_d, because a reproduction cannot meaningfully be shown to agree with a published curve to better than the precision with which that curve can be read.

The parameters were taken exactly as printed in the anchor and are listed in table 2. No parameter was fitted, tuned or adjusted at any point. The digitisation mask was refined during the extraction, since an initial legend mask clipped the upper curve band; correcting the mask boundary improved the agreement between the two independent passes by a factor of about twenty five, and the corrected extraction is used throughout.

Table 2: Parameters used for the reproduction, taken verbatim from the anchor [26]. Dimensionless throughout, with $\tau_\Delta = 0.04$ in every case. No value was fitted.

Anchor figure	**Regime**	τ_q	κ^2	**Series terms / time span**
Fig. 2	convergence study	0.02	0.02	$N = 1, 3, 10, 40$; $t \in [0, 1]$
Fig. 3	Fourier resonance	0.02	0.02	$N = 40$; $t \in [0, 1]$
Fig. 5	over-diffusive	0.02	0.20	$N = 10$; $t \in [0, 2]$

5.3. Reproduction accuracy

The first reference case is presented in fig. 3 and the second in fig. 4; the metrics for all three cases are collected in table 3. Each reproduction uses the published parameters directly, with no fitting. The three reference cases are three figures of a single published study [26] rather than three independent publications, and are described as such throughout.

The outcome is reported in table 3, with the overlays and residuals in fig. 3. For the two solution figures the reproduction error falls below the digitisation noise floor: the residual is dominated by the precision with which the published curves can be read, not by any discrepancy between the published solution and the present implementation, with the great majority of residuals falling within $\pm 2\sigma_d$ in both cases.

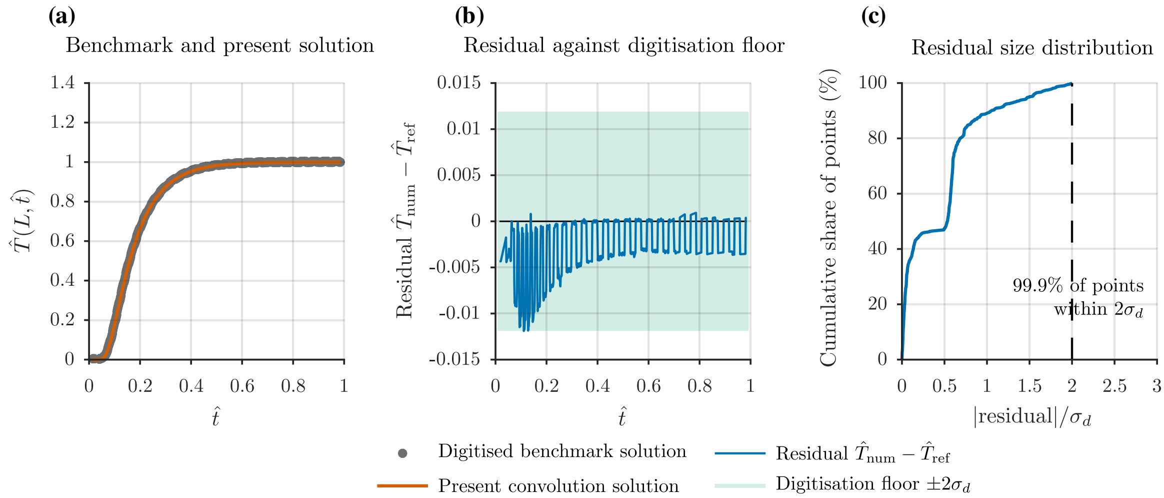

Figure 3: External validation of the forward operator against the first reference case, the resonance condition $B = 1$. (a) Digitised benchmark solution read from the published figure, overlaid with the present convolution solution evaluated at the literature parameters with no fitting. (b) Residual $\hat{T}_{\text{num}} - \hat{T}_{\text{ref}}$ against the digitisation floor $\pm 2\sigma_d$, where σ_d is the standard deviation attributable to reading the published curve. (c) Empirical cumulative distribution of $|\text{residual}|/\sigma_d$, with the bound $2\sigma_d$ marked. Verified metrics: NRMSE = 0.384%, RMSE = 3.84×10^{-3}, $n = 729$ comparison points, and 99.9% of residuals within $\pm 2\sigma_d$. External validation of the forward operator only; not validation of the inverse conclusions and not experimental validation.

The convergence figure of the anchor, which shows the eigenfunction expansion truncated at successively larger numbers of terms, is reproduced with a pooled normalised error of 0.483%. One entry requires comment. The $N = 40$ curve of that figure is nearly converged and spans a temperature range of only 0.031, so normalising its error by its own range gives 8.19% while the absolute error, 2.7×10^{-3}, is comparable to the other curves. Normalised against the full range of the figure the same residual is 0.162%. Both

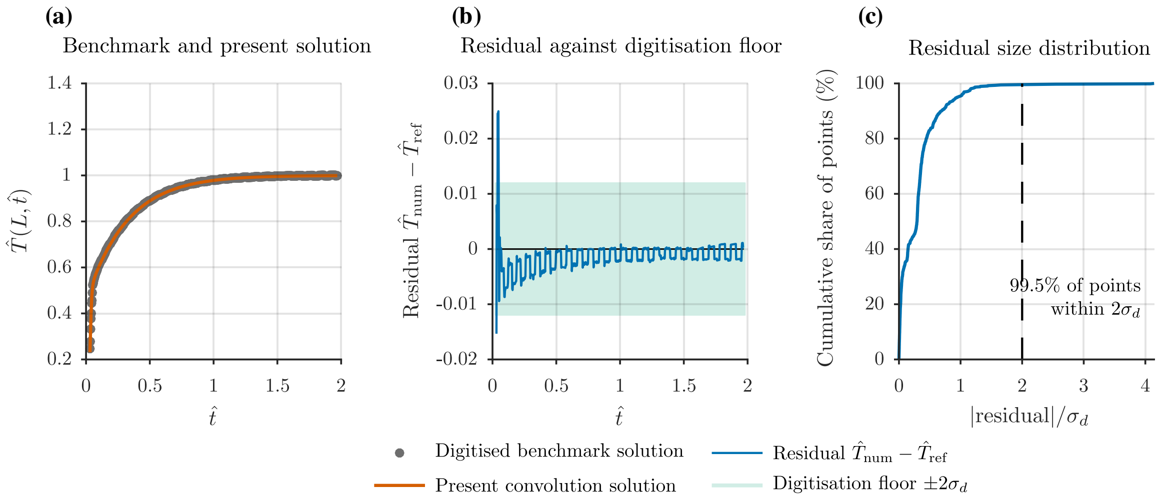

Figure 4: External validation of the forward operator against the second reference case, the over-diffusive condition, using the same three-panel construction and the identical unmodified solver as fig. 3. Panels are as in fig. 3: (a) digitised benchmark solution and present convolution solution, (b) residual against the digitisation floor $\pm 2\sigma_d$, (c) empirical cumulative distribution of the scaled residual. Verified metrics: NRMSE $= 0.382\%$, RMSE $= 2.88 \times 10^{-3}$, $n = 743$, and 99.5% of residuals within $\pm 2\sigma_d$. The three reference cases used in this work are three figures of the single published study [26], not three independent publications.

numbers are reported in table 3; the large value is an artefact of normalising by a small range, not a failure of reproduction.

Table 3: External validation of the forward operator against the published figures of the anchor [26]. σ_d is the digitisation noise floor (41). A ratio RMSE/$\sigma_d < 1$ means the reproduction error is below the precision with which the published curve can be read. These figures are reproductions for validation, not original results.

Figure	**Points**	**RMSE**	**NRMSE**	$E_{\max}$	σ_d	**RMSE/σ_d**
Fig. 3 (resonance)	729	0.00384	0.384 %	0.01189	0.00593	0.65
Fig. 5 (over-diffusive)	743	0.00288	0.382 %	0.02496	0.00603	0.48
Fig. 2 (pooled)	1999	0.00819	0.483 %	0.05813	—	—
$N = 40$ subset, own range	109	0.00275	8.19 %	0.01044	—	—
$N = 40$ subset, full range	109	0.00275	0.162 %	0.01044	—	—

Entries marked — are not applicable: the pooled convergence figure reports curves at several truncation levels rather than a single digitised trace, so a digitisation standard deviation σ_d is not defined for it.

The residuals show a low amplitude sawtooth of about one pixel, consistent with pixel quantisation of the source image, and the largest pointwise deviations occur where the published curve rises steeply, where a small horizontal digitisation error maps to a large vertical one. Neither feature indicates a model discrepancy.

5.4. Scope established by the reproduction

The forward framework reproduces three published figures of an independently authored analytical solution, using the published parameters without adjustment, to within the precision with which those figures can be read. The forward operator underlying every subsequent computation in this paper is therefore externally validated. The inverse problem is not validated by this exercise, and is not claimed to be.

6. Numerical Certification

The numerical framework of section 4 resolves the quantities reported in this paper far above its own error floor, as established below. The agreement between solvers reported here is numerical cross-checking, not external validation.

6.1. *Failure mode of the Laplace inversion*

The most consequential result of the certification concerns the numerical Laplace inversion, for which a direct implementation of the transform admits a parity-dependent failure mode that is not signalled by any convergence diagnostic.

The effect is quantified in fig. 5. If the transform (23) is inverted by applying the fixed Talbot rule to the material kernel and handling the pulse factor $1 - \mathrm{e}^{-s\tau_\Delta}$ by the second shift theorem, the computed solution is catastrophically wrong at one isolated time, while remaining accurate everywhere else. The cause is exact and predictable. The pulse

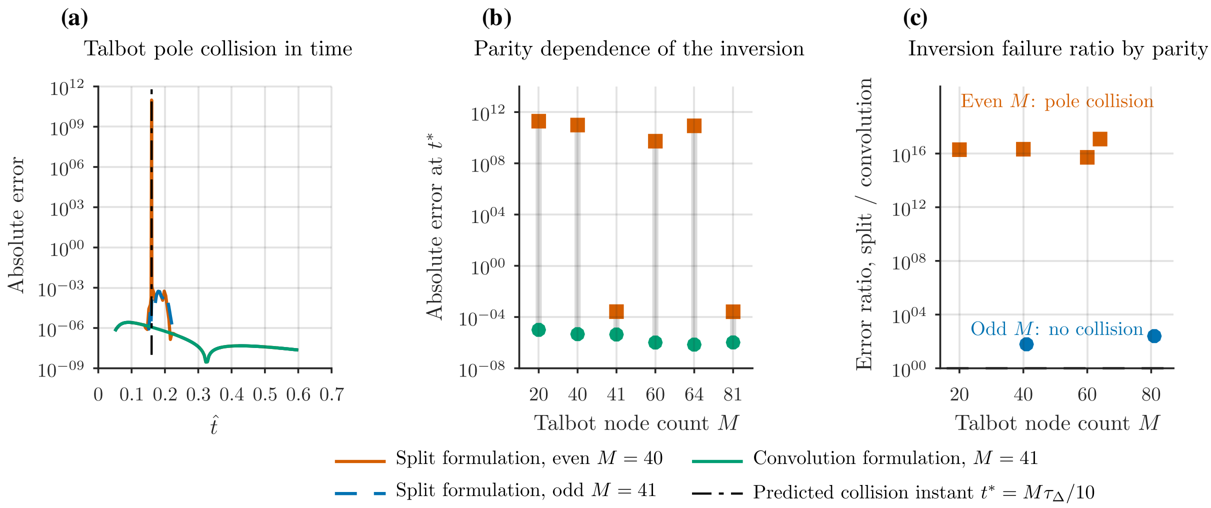

Figure 5: Certification of the numerical Laplace inversion. (a) Absolute error against time for the split formulation at even node count $M = 40$ and odd node count $M = 41$, and for the adopted convolution formulation, with the predicted pole-collision instant $t^* = M\tau_\Delta/10$ marked. (b) Absolute error at $t = t^*$ for six node counts, comparing the split and convolution formulations; the connecting segments indicate the discrepancy at each M. (c) Ratio of the split to the convolution error, separating even from odd node counts. The split formulation fails by up to 9.7×10^{10} at even M, whereas the convolution formulation remains at $\mathcal{O}(10^{-6})$ throughout. The failure is a property of the numerical inversion and not of the physics: the exact solution is regular at t^*. The convolution formulation with $M = 41$ is used throughout this work.

transform (24) has removable singularities at $s = \pm\mathrm{i}\omega$, $\omega = 2\pi/\tau_\Delta$, which are cancelled by the zeros of $1 - \mathrm{e}^{-s\tau_\Delta}$. Splitting the two factors and inverting $\omega^2/[s(s^2 + \omega^2)]$ alone reinstates genuine poles at $s = \pm\mathrm{i}\omega$. The Talbot contour (39) meets the imaginary axis only at $\theta = \pi/2$, which is a quadrature node exactly when M is even, and there $s = \mathrm{i}r\pi/2$. Setting $r\pi/2 = \omega$ with $r = 2M/(5t)$ gives the failure time

$$t^* = \frac{M\,\tau_\Delta}{10}. \tag{42}$$

This prediction was tested directly. For $\tau_\Delta = 0.04$ and even $M \in \{20, 40, 60, 64\}$ the error at t^* is of order 10^9 to 10^{11}, while at $0.98\,t^*$ and $1.02\,t^*$ it remains of order 10^{-4}; for odd $M = 41$ no such spike occurs at any time. The two remedies are therefore to use an odd number of nodes, which removes the node from the imaginary axis, and better, to avoid the splitting altogether. The convolution formulation (38) adopted in section 4.1 does the latter: it inverts only the pole free kernel $\hat{K}$, so the removable singularities never appear.

Two points follow. First, this is a property of the numerical inversion, not of the physics: the exact solution is perfectly regular at t^*, and nothing physical happens there. Second, the usual remedy of increasing the number of quadrature nodes is not merely ineffective here but actively harmful, since t^* scales with M and simply moves the failure to a later time while round off grows. With the convolution formulation the accuracy plateau extends across $M \in [25, 81]$, degrading beyond $M \approx 91$ through round off; $M = 41$ is used throughout.

6.2. Spatial convergence and arithmetic precision

Figure 6 collects the convergence evidence. Panels (a) and (b) quantify the spatial discretisation error and its observed order, panel (c) the agreement between the truncated eigenfunction series and the convolution solution, and panel (d) the correspondence between the sampling and asymptotic uncertainty estimates. All four are internal verification of the present implementation; external validation is reported separately in section 5.

With the corrected formulation the certification results are as follows. Against the arbitrary precision reference of section 4.3, the convolution solver agrees to between 10^{-11} and 10^{-13} at times away from the steep rise, and to about 10^{-7} against the independent finite difference solver across the whole record, the latter being limited by the finite difference discretisation rather than by the inversion. Increasing the Gauss–Legendre order from 32 to 64 changes the result by at most 10^{-13}, so the convolution quadrature is fully converged.

The finite-difference solver converges at the expected second order in space. Measured against a Richardson reference the observed order is 2.00 and the spatial error at the production resolution $N_x = 400$ is 2.6×10^{-6}; measured against the independent $N_x = 3200$ solution of fig. 6(a) the corresponding error is 1.00×10^{-5} and the per-interval orders are 2.00, 2.02, 2.07 and 2.32. The two figures differ because they use different reference solutions, both retained as independent checks.

The temporal error at relative tolerance 10^{-10} is of order 10^{-9}. Repeating the arbitrary precision evaluation at 15, 30 and 50 digits changes the result by no more than 8.9×10^{-16}, confirming that double precision is sufficient and that none of the reported quantities is limited by round off.

The eigenfunction series used by the anchor converges as $O(N^{-2})$, the ratio of successive errors tending to 4.0; at $N = 40$ the truncation error is 3.1×10^{-3}, and $N \geq 80$ is required to reach 7.8×10^{-4}. This matters only for reproducing the anchor's own truncated curves in section 5, not for the present solvers.

6.3. Limiting cases and physical admissibility

The limits of section 2.7 were checked numerically. Setting $\kappa^2 = \alpha\tau_q$ reproduces the analytical Fourier solution to 2.6×10^{-6}, an independent numerical confirmation of proposition 1. Letting $\tau_q, \kappa^2 \to 0$ likewise recovers the Fourier solution to 2.6×10^{-6}. The computed temperature remains non negative throughout in every case tested, in agreement with the anchor and in contrast to the negative temperature artefacts reported for dual

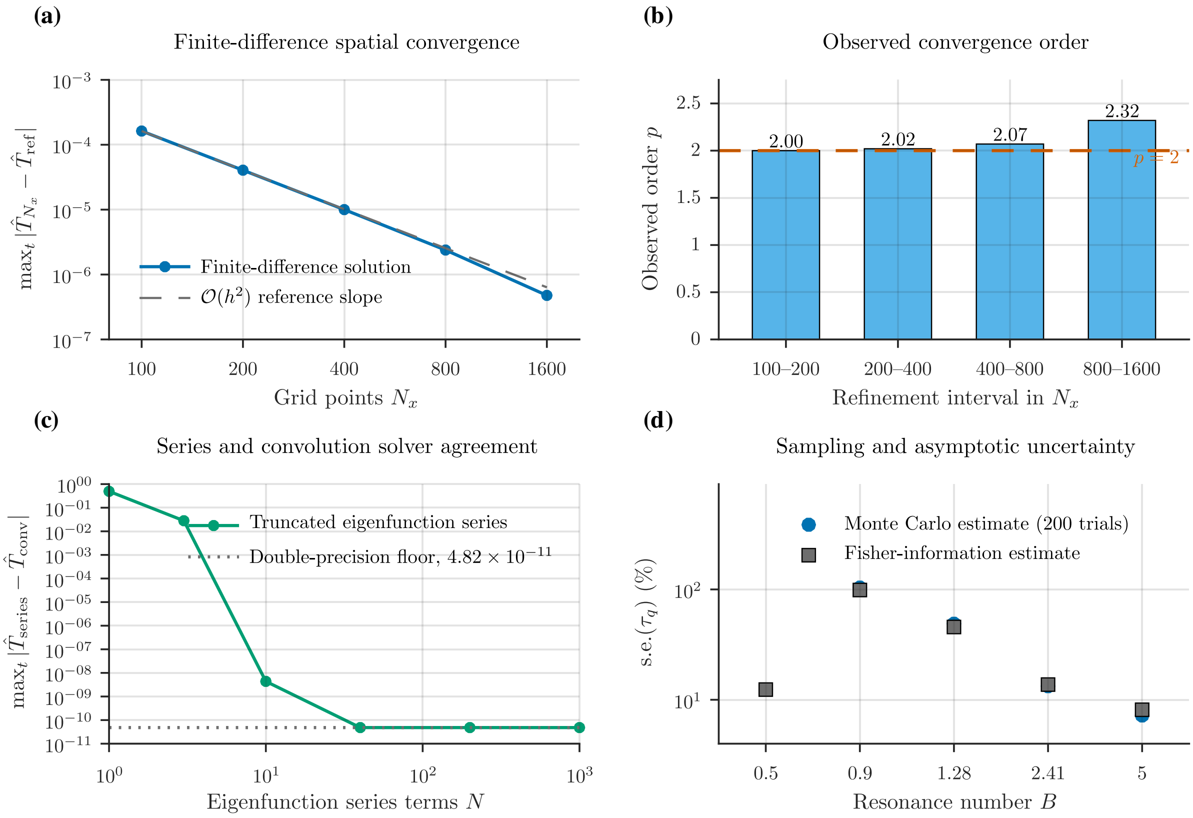

Figure 6: Numerical verification of the solver and characterisation of the inverse procedure, assembled from four complementary lines of evidence. (a) Finite-difference spatial error against the grid resolution N_x on logarithmic axes, measured against the $N_x = 3200$ reference solution, with a second-order reference slope. (b) Observed convergence order p for each successive refinement interval: 2.00, 2.02, 2.07 and 2.32, against the theoretical value $p = 2$. These four values originate from the documented Windows verification run and are recorded there, and reproduced here, to four significant figures; a machine-readable subset of the same refinement ladder ($N_x = 100$ to 400), independently regenerable from the supplied pipeline, gives orders of 2.02 and 2.07 for its two intervals, confirming the same near-second-order behaviour without reproducing the four-value ladder digit for digit. (c) Agreement between the truncated eigenfunction series and the convolution solution against the number of retained terms N, descending to the double-precision floor 4.82×10^{-11}. (d) Monte Carlo estimate of s.e.(τ_q) from 200 trials against the asymptotic Fisher-information estimate at each resonance number. Panels (a)–(c) verify the numerical convergence of the present implementation and are internal cross-checking rather than external validation; panel (d) characterises the behaviour of the inverse procedure on controlled synthetic data. No panel constitutes external or experimental validation.

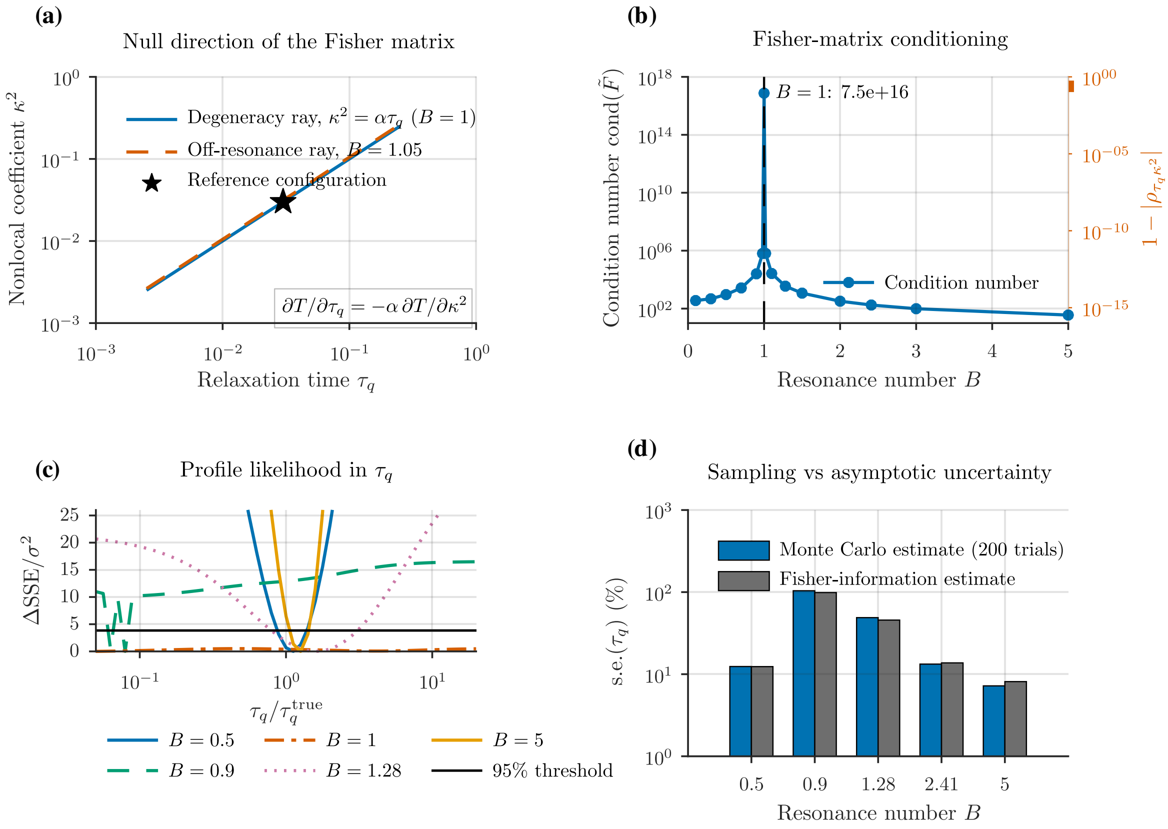

Figure 7: Structural and practical non-identifiability of the (τ_q, κ^2) split at the resonance condition. (a) Null direction of the Fisher matrix: the analytic degeneracy ray $\kappa^2 = \alpha\tau_q$ $(B = 1)$ along which the observable is invariant, the off-resonance ray at $B = 1.05$, and the reference configuration, with the exact identity $\partial T/\partial\tau_q = -\alpha\,\partial T/\partial\kappa^2$ annotated. (b) Condition number of the Fisher matrix against B on a logarithmic ordinate, attaining 7.50×10^{16} at $B = 1$, together with the correlation deficit $1 - |\rho_{\tau_q\kappa^2}|$ on the right-hand axis, which collapses to 3.3×10^{-16} at the same point. (c) Profile likelihood in τ_q at each computed resonance number, with the 95% threshold from the χ^2_1 distribution; the $B = 0.9$ curve is multimodal because the re-optimisation at fixed τ_q reaches different local minima at adjacent grid nodes, and no interval is read from it. (d) Monte Carlo estimate of s.e.(τ_q) from 200 trials against the asymptotic Fisher-information estimate, at each computed resonance number.

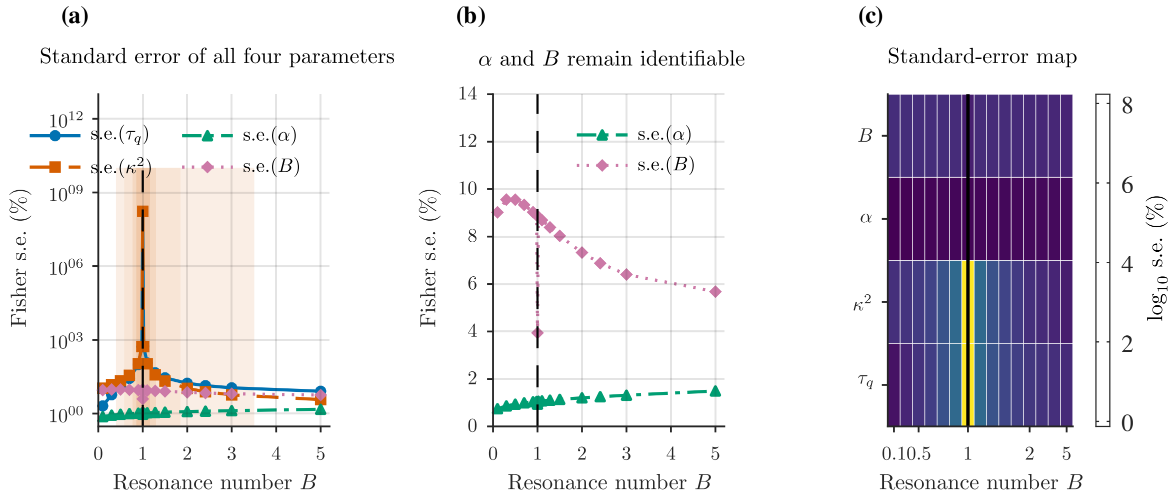

Figure 8: Practical identifiability as a function of the resonance number. (a) Fisher standard errors of τ_q, κ^2, α and B against B on a logarithmic ordinate. The shaded regions are the verified bands within which s.e.(τ_q) exceeds 10, 20, 50 and 100%; the 20% band is $B \in [0.628, 1.804]$. (b) The same standard errors for α and B on a linear ordinate, showing that both remain small and nearly flat through resonance, α to approximately 1% and B to between 4 and 10%. (c) Map of $\log_{10}$ s.e. for all four parameters against B, with the resonance column marked. Standard errors are asymptotic and correspond to the configuration of section 7.1.

phase lag models [57], and the long time limit satisfies $T(1, t \to \infty) = 1.00000000$ as required by the normalisation (10).

6.4. Error budget

Table C.1 assembles the error budget. The total numerical error of the framework is of order 10^{-6}, whereas the digitisation noise floor of the external validation exercise is 6×10^{-3}, three orders of magnitude larger. The external validation of section 5 is therefore digitisation limited, not solver limited: the agreement reported there is as close as the published figures permit.

7. Practical Identifiability

section 3 established an exact degeneracy at the single point $B = 1$. A point degeneracy would be of limited practical interest if the estimation problem recovered quickly on either side of it. It does not: the degeneracy extends into a wide neighbourhood within which τ_q cannot be estimated to any useful precision. The quantities that remain determined in that neighbourhood are identified below.

7.1. Reference configuration

All results in this section are for one stated configuration: $\tau_q = 0.03$, $\tau_\Delta = 0.04$, 80 observation times uniformly spaced on $t \in [0.02, 1.2]$, independent Gaussian noise of 2% of the temperature range, estimation in log parameters with the analytic Jacobian of section 4.4. Working units are $L = 1$, $\alpha = 1$, $t_p = 0.04$. The numerical values of the band boundaries reported below depend on this configuration and are not universal constants. Only the singularity at $B = 1$ is configuration independent, because it is structural.

7.2. Fisher-information conditioning

Figure 7 assembles the diagnostics of the same structural result, and fig. 8 maps the standard errors and conditioning across B. The Fisher conditioning, the profile likelihood and the Monte Carlo study are methodologically independent: the first uses analytic derivatives at a point, the second re-optimises the objective along a fixed direction, and the third samples the estimator itself. Their agreement indicates that the degeneracy is a property of the parameter-to-observable map rather than an artefact of any single statistical construction. The claim remains confined to the (τ_q, κ^2) split, since α and B stay identifiable throughout.

Table 4 and fig. 8 report the Fisher analysis as a function of B. The behaviour is a smooth deterioration towards the resonance rather than an isolated pathology. Far from resonance the problem is well conditioned: at $B = 0.10$ the condition number of the Fisher matrix is 3.6×10^2 and the standard error on τ_q is 2.05%. As $B \to 1$ the condition number rises by fourteen orders of magnitude, the estimator correlation between τ_q and κ^2 tends to $+1$, and the standard error on τ_q diverges. At $B = 1$ the Fisher matrix is numerically singular, as proposition 2 requires.

The mechanism is the progressive alignment of the two sensitivity directions. Away from resonance the response retains a component that distinguishes a change in relaxation from a change in nonlocality, since the factor $(1 + \tau_q s)$ in the numerator of (21) and the factor $(\alpha + \kappa^2 s)$ in its denominator act on different regions of the transform variable. As $B \to 1$ these two factors approach proportionality and cancel in the limit, so a perturbation in τ_q becomes reproducible by a compensating perturbation in κ^2. The observable consequence is that the rise portion of the rear-face history, which carries most of the information about the relaxation mechanism, becomes insensitive to the split while remaining fully sensitive to α. This is why the diffusivity and the resonance number stay well determined while the individual coefficients do not.

Two correlation coefficients occur in this problem, with opposite signs. The coefficient tabulated in table 4, and plotted in fig. 7(b), is the estimator correlation

$$\rho_{\tau_q \kappa^2} = [F^{-1}]_{12} \Big/ \sqrt{[F^{-1}]_{11}[F^{-1}]_{22}}$$

obtained from the inverse Fisher matrix, which is the correlation of the sampling distribution of the estimates and tends to $+1$ at resonance: an overestimate of τ_q is compensated by an overestimate of κ^2 along the ray $\kappa^2 = \alpha\tau_q$. The coefficient -1 of proposition 2 is the normalised off-diagonal of the Fisher matrix itself, $F_{12}/\sqrt{F_{11}F_{22}}$, which follows from the anti-parallel sensitivities (32). Both express the same rank-one degeneracy with null direction $(1, \alpha)$; the sign difference is a property of inverting a singular two-by-two block and not a discrepancy.

The Monte Carlo results in the same table are consistent with the Fisher predictions where the latter are valid. At $B = 0.50$ the Fisher standard error on τ_q is 12.4% against a Monte Carlo value of 12.4%; at $B = 0.90$, 98.8% against 103.9%; at $B = 1.28$, 45.6% against 48.9%; at $B = 2.41$, 13.7% against 13.3%; at $B = 5.0$, 8.1% against 7.2%. The two estimates agree throughout the range in which the asymptotic approximation is meaningful, which supports the use of the Fisher standard error as a summary statistic outside the immediate neighbourhood of the resonance. The agreement is contingent on analytic sensitivities: Jacobians formed by differencing the forward solver at its own noise floor produce a spurious discrepancy between the two estimates, which is why the analytic route of section 4.4 is used throughout.

Near resonance the two diverge for a well understood reason: the log likelihood ceases to be quadratic, so the asymptotic Fisher standard error is no longer a meaningful summary, and the Monte Carlo sampling distribution acquires heavy tails for which a standard deviation is dominated by a few realisations. In that regime the percentile ranges of table 4 are the informative statistic, and the tabulated Monte Carlo standard error is computed robustly from the interquartile range. At $B = 0.90$ the central 90% of Monte Carlo estimates of τ_q spans a factor of about 65; at $B = 1.00$ the estimates are unbounded, as they must be.

Table 4: Practical identifiability as a function of B for the configuration of section 7.1: $\tau_q = 0.03$, $\tau_\Delta = 0.04$, 80 points on $t \in [0.02, 1.2]$, 2% noise, log-parameters, analytic Jacobians. Fisher standard errors are asymptotic; Monte Carlo uses 200 noise realisations started at the true parameters, summarised by the interquartile range divided by 1.349 because the sampling distribution is heavy tailed near resonance. Percentile ranges are stated as multiples of the true value. The numerical boundaries are configuration dependent; the singularity at $B = 1$ is not.

	Fisher analysis					**Monte Carlo**	
B	cond(F)	corr(τ_q, κ^2)	s.e. τ_q	s.e. α	s.e. B	s.e. τ_q	τ_q 5–95%
0.10	3.6×10^2	+0.767	2.05 %	0.75 %	9.02 %	—	—
0.30	4.8×10^2	+0.926	5.83 %	0.86 %	9.56 %	—	—
0.50	9.3×10^2	+0.974	12.37 %	0.94 %	9.56 %	12.4 %	0.84–1.23
0.70	2.7×10^3	+0.993	27.06 %	0.99 %	9.34 %	—	—
0.90	2.5×10^4	+0.999	98.75 %	1.04 %	9.03 %	103.9 %	0.09–5.88
0.98	6.4×10^5	+1.000	526.6 %	1.06 %	8.90 %	—	—
1.00	$\mathbf{7.5 \times 10^{16}}$	**singular**	**divergent**	0.94 %	4.00 %	unbounded	unbounded
1.02	6.5×10^5	+1.000	542.5 %	1.06 %	8.83 %	—	—
1.10	2.7×10^4	+0.999	114.7 %	1.08 %	8.69 %	—	—
1.28	3.5×10^3	+0.996	45.64 %	1.11 %	8.39 %	48.9 %	0.35–1.85
1.50	1.2×10^3	+0.987	28.49 %	1.14 %	8.03 %	—	—
2.00	3.2×10^2	+0.957	17.19 %	1.21 %	7.33 %	—	—
2.41	1.8×10^2	+0.927	13.73 %	1.25 %	6.88 %	13.3 %	0.79–1.20
3.00	9.8×10^1	+0.882	11.17 %	1.32 %	6.40 %	—	—
5.00	3.6×10^1	+0.757	8.12 %	1.49 %	5.68 %	7.2 %	0.88–1.12

7.3. Profile likelihood

The Fisher analysis assumes a quadratic log likelihood, which fails precisely where the problem is most interesting. Profile likelihood does not make that assumption. Fixing τ_q on a grid and re-optimising α and κ^2 at each node gives the intervals of fig. 9, obtained at the $\chi^2(1, 0.95) = 3.841$ threshold.

The interval widths track the conditioning. At $B = 5.0$ the 95% interval for τ_q spans a factor of 1.26, and at $B = 0.50$ a factor of 1.41. At $B = 1.28$, the value of the refined basalt calibrations, it spans a factor of 3.55, from 0.79 to 2.82 times the true value. At $B = 1.00$ the profile remains below the threshold across the entire grid, a factor of 398 from 0.05 to 19.95 times the true value, so the interval is bounded only by the grid and is unbounded in fact, as proposition 2 requires.

The row at $B = 0.90$ is reported separately because it is not a clean profile. The re-optimisation over (α, κ^2) at fixed τ_q converges to different local minima at adjacent grid nodes, so the computed curve is not unimodal and its lowest point falls at 0.08 times the true relaxation time rather than near it. A width factor read off such a curve would not be a confidence interval, and none is quoted here. The failure is itself diagnostic: at $B = 0.90$ the objective in the (τ_q, κ^2) plane is already flat enough, over more than an order of magnitude in τ_q, that a two-start local optimiser does not reliably locate the conditional optimum. The corresponding Fisher and Monte Carlo standard errors at this condition,

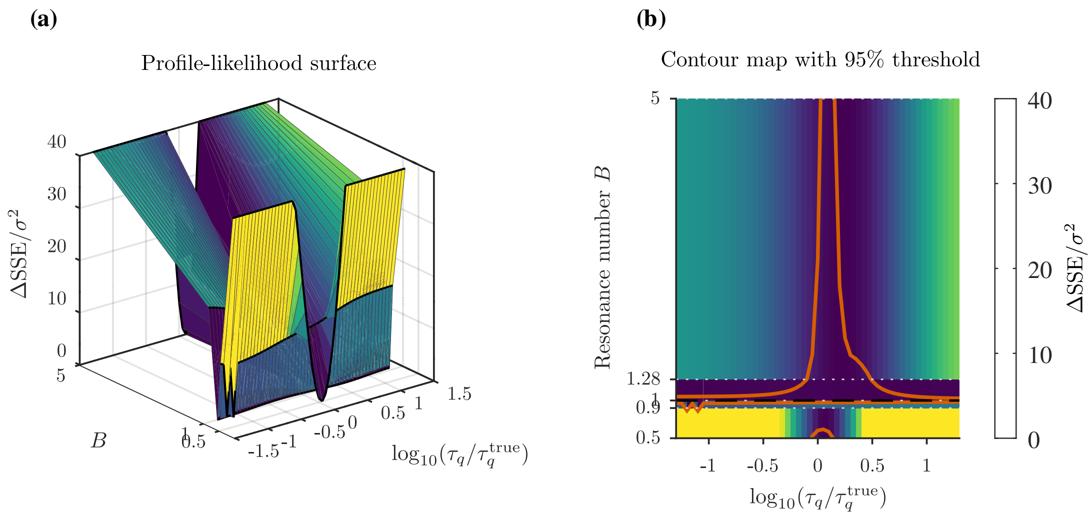

Figure 9: Profile likelihood over the parameter plane (B, τ_q). (a) Three-dimensional profile-likelihood surface of $\Delta\mathrm{SSE}/\sigma^2$ over the 265 computed points, five values of B by 53 values of τ_q. The resonance number is drawn on its true numerical axis, the black curves mark the five computed rows, and no value is interpolated in B; the surface is clipped at 40 for display only. (b) The same data as a contour map with the 95% threshold $\chi^2_1 = 3.841$ marked. The admissible interval in τ_q widens without bound as $B \to 1$: the width factor is 1.41 at $B = 0.5$, 3.55 at $B = 1.28$ and 1.26 at $B = 5$, while at $B = 1$ the profile stays below the threshold across the whole grid. The row at $B = 0.9$ is multimodal, because the re-optimisation at fixed τ_q reaches different local minima at adjacent nodes, and no interval is quoted from it.

98.8% and 103.9%, are obtained without re-optimisation and are unaffected. The profile rows at the remaining four conditions are unimodal and are the ones used in the argument.

The factor of 3.55 at $B = 1.28$ is the conditioning of a real calibration, and it brackets the factor of 1.51 to 3.50 by which two published evaluations of the same basalt specimens disagree, as discussed in section 9.4.

7.4. Width of the ill-conditioned band

Inverting the Fisher standard error as a function of B gives the region within which τ_q cannot be recovered to a stated precision:

$$\mathrm{s.e.}(\tau_q) > 10\% : \quad B \in [0.441,\ 3.464], \tag{43}$$
$$\mathrm{s.e.}(\tau_q) > 20\% : \quad B \in [0.628,\ 1.804], \tag{44}$$
$$\mathrm{s.e.}(\tau_q) > 50\% : \quad B \in [0.817,\ 1.252], \tag{45}$$
$$\mathrm{s.e.}(\tau_q) > 100\% : \quad B \in [0.901,\ 1.116]. \tag{46}$$

The exact degeneracy at a point therefore expands into a band of substantial width. At the 20% criterion the relaxation time is not usefully recoverable anywhere in $B \in [0.628, 1.804]$, a range spanning nearly a factor of three. These boundaries are properties of the configuration of section 7.1; the singularity at their centre is not.

7.5. Well-determined parameter combinations

The preceding results are negative statements about τ_q and κ^2 individually. They are not statements about the informativeness of the data, and it would be wrong to conclude that a measurement near resonance is worthless. Table 4 also reports the standard errors of α and of B in the parameterisation (30). Across the whole range, including exactly at

$B = 1$, the diffusivity is determined to about 1% and the resonance number to between 4% and 10%. The information content of the record is essentially undiminished near resonance; what fails is the resolution of a well determined combination into its two components.

This is the constructive form of the result. Reporting (α, B), with τ_q and κ^2 quoted jointly rather than as independent point estimates, is well posed everywhere. Reporting τ_q and κ^2 as separate material constants is not well posed for a material whose B lies in the band (44). The recommendation of section 10 follows directly.

7.6. Optimiser dependence

Because the objective develops a long flat valley near resonance, the estimate may depend on where the optimiser is started; this dependence is real but, in the tests reported in Appendix D, smaller than the effect of measurement noise itself, motivating multi-start or global search strategies in poorly conditioned regimes.

8. Model Selection

The identifiability result can be reached by a second, statistically independent route. If at $B = 1$ the Guyer–Krumhansl response is observationally a Fourier response, then a model selection criterion applied to Guyer–Krumhansl data at resonance should prefer the Fourier model, because the two extra parameters purchase no improvement in fit while incurring a complexity penalty. That prediction is tested below.

8.1. Candidate models and protocol

The four candidates are exact special cases of the single transfer function (21), obtained by constraining parameters as derived in section 2.7: Fourier ($\tau_q = 0$, $\kappa^2 = 0$, $k = 1$ parameter), the Maxwell–Cattaneo–Vernotte model ($\kappa^2 = 0$, $k = 2$), the Nyíri model ($\tau_q = 0$, $k = 2$) and the full Guyer–Krumhansl model ($k = 3$). Because all four share one transfer function, they also share the solver, the estimation procedure, the convergence criteria and the noise model; no model receives individual tuning. This is what makes the comparison like for like.

The Maxwell–Cattaneo–Vernotte and Nyíri models are included because they isolate the two mechanisms of the Guyer–Krumhansl equation separately, so the comparison tests which mechanism the data require rather than merely whether more parameters fit better. Models requiring a state variable absent from (21), such as those introducing a thermal displacement, are excluded: they cannot be expressed as constraints on (21) and including them would break the like for like protocol. This exclusion is stated rather than passed over silently.

Selection uses the Bayesian information criterion of section 4.5, averaged over 15 noise realisations at each condition, with data generated by the full Guyer–Krumhansl model at 2% noise. The Bayesian information criterion is used in preference to the Akaike information criterion because its stronger parsimony penalty gives the more conservative demonstration that the rank deficiency of proposition 2 is recoverable from a selection statistic alone, without reference to any derivative or Fisher-matrix information.

8.2. Information-criterion scores

The outcome is reported in table 5 and fig. 10. Away from resonance the Guyer–Krumhansl model is selected decisively, and increasingly so as B moves away from unity: at $B = 5.0$ the Fourier model is penalised by 267 BIC units. At $B = 1.00$ the ordering reverses and the Fourier model is selected, with the Guyer–Krumhansl model penalised by 6.64 units despite the data having been generated by the Guyer–Krumhansl model itself. The reversal is the expected behaviour of the criterion rather than a limitation of it.

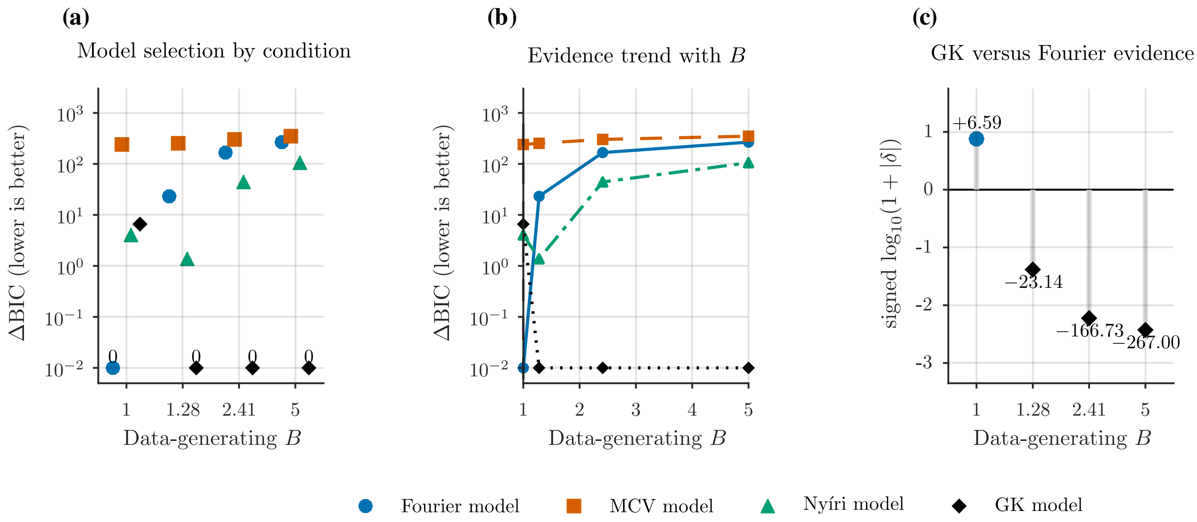

Figure 10: Like-for-like model comparison over the four nested candidates, each an exact special case of the single transfer function: the Fourier model ($k = 1$ parameter), the Maxwell–Cattaneo–Vernotte model ($k = 2$), the Nyíri model ($k = 2$) and the Guyer–Krumhansl model ($k = 3$). The criterion ΔBIC is referenced to the best-scoring model at each condition, so lower values indicate stronger support and the selected model scores zero. (a) ΔBIC by data-generating condition on a logarithmic ordinate; exact zeros are annotated explicitly because a logarithmic axis cannot represent zero. (b) Trend of the evidence with B. (c) Signed difference $\delta = \Delta\mathrm{BIC}_{\mathrm{GK}} - \Delta\mathrm{BIC}_{\mathrm{Fourier}}$, plotted as $\mathrm{sign}(\delta)\log_{10}(1+|\delta|)$ so that all four conditions remain visible across three orders of magnitude, with the exact values printed. A positive value indicates that the Fourier model is preferred even though the data were generated by the Guyer–Krumhansl model, which occurs at $B = 1$. Synthetic data at 2% noise, 15 realisations per condition, identical estimation protocol for every candidate.

At resonance the two models make identical predictions, by proposition 1, so the additional parameters reduce the residual only by fitting noise and the parsimony penalty rejects them. The rank deficiency of proposition 2 is thereby recovered by a route that uses no derivative information and no Fisher matrix.

The Maxwell–Cattaneo–Vernotte model is rejected everywhere by a wide margin, 237 to 349 BIC units, consistent with the finding of Fehér and Kovács that this model cannot describe such experiments [27]. The Nyíri model, which retains the nonlocal term and discards relaxation, performs far better than the Maxwell–Cattaneo–Vernotte model and is only marginally worse than the full model near resonance. The ordering indicates that near $B = 1$ the data do not require the relaxation mechanism.

8.3. Statistical scope of the selection result

The Bayesian information criterion measures statistical parsimony under the specified data generating process, noise model and observation design. It is used here as an identifiability diagnostic and not as evidence about which constitutive theory is physically correct. That the criterion prefers Fourier at $B = 1$ is a statement about what the data can distinguish, not a statement that the material obeys Fourier's law. This distinction matters when interpreting model selection applied to real measurements: a Fourier preference for a specimen near resonance is uninformative about the underlying transport physics.

Table 5: Bayesian information criterion differences relative to the best model at each condition, averaged over 15 noise realisations. Data generated by the Guyer–Krumhansl model at 2% noise; all four candidates are special cases of (21) fitted with an identical protocol. Lower is better; 0.00 marks the selected model. The values are those of the frozen reference calculation. Four of the sixteen tabulated entries – MCV and GK at $B = 1.00$ (236.58 and 6.64) and MCV at $B = 1.28$ and $B = 5.00$ (253.21 and 349.25) – are reproduced by independent re-evaluation with the production convolution solver during cross-platform validation as 240.31 and 6.59 at $B = 1.00$ and as 253.63 and 349.26 at $B = 1.28$ and $B = 5.00$ respectively; the remaining twelve of sixteen entries agree to the digits shown. The differences arise in the two- and three-parameter fits at $\kappa^2 \to 0$, where the two implementations of (21) differ by $O(10^{-12})$ and the Levenberg–Marquardt iteration amplifies that difference into the reported sum of squares. No model ranking, selection or conclusion is affected.

True B	Fourier	MCV	Nyíri	GK	Selected
1.00	**0.00**	236.58	4.03	6.64	**Fourier**
1.28	23.14	253.21	1.38	**0.00**	GK
2.41	166.73	302.12	44.21	**0.00**	GK
5.00	267.00	349.25	105.29	**0.00**	GK

9. Implications for Published Calibrations

The preceding sections are statements about a model. The location of published Guyer–Krumhansl calibrations relative to the identified ill-conditioned regime is evaluated below, together with the consequences for the interpretation of the reported values.

9.1. Scope of the calibration audit

The interpretation of the aggregate calibration statistics requires consideration of the admissible B-domain used in the source literature. Fehér and Kovács state that their evaluation is restricted to the domain $1 \leq B < 3$, and identify its lower limit with the Fourier case [27]. The accepted evaluation procedure therefore constrains B to an interval whose lower boundary is the exact degeneracy analysed in section 3. Values of B close to unity in the literature are in part a consequence of that admissible domain and of the physical expectation, well supported by the measurements, that these materials behave over diffusively. They reflect a structural property of the estimation problem, not an incidental or coincidental property of the materials. The restriction is well motivated: bounding a nonlinear search improves its stability, and the domain reflects the observed regime.

9.2. Audited calibrations

Table 6 lists all twelve Guyer–Krumhansl heat pulse calibrations reported in [8, 27] for which the published parameters are sufficient to recompute B from (16); no case meeting this criterion was excluded, and no broader literature search beyond these two sources was undertaken. Every cell is flagged as reported by the original authors (R) or calculated in the present study (C). Where the underlying parameters are tabulated, B was recomputed from (16); the recomputed values agree with the authors' own reported values to within 0.016 in every case where both exist, the largest deviation being for basalt at 3.84 mm. For the six basalt entries both the parameters and the resulting B were read directly from Tables 1 and 2 of [27], which report the earlier and the refined evaluations side by side.

Eight of the twelve calibrations have B inside the band (44) within which the relaxation time carries a standard error exceeding 20% under the tested configuration, as shown in fig. 11(a): all six basalt evaluations, which cluster at $B = 0.94$ to 1.31, the leucocratic

Table 6: Audited Guyer–Krumhansl calibrations. R denotes a value reported by the original authors; C denotes a value calculated in the present study from the reported parameters using (16). The band is (44), within which s.e.$(\tau_q) > 20\%$ for the configuration of section 7.1. Eleven of the twelve sources report no parameter uncertainty; case 12 is the exception.

#	Source	Material	α $[10^{-6}\,\mathrm{m^2\,s^{-1}}]$	τ_q [s]	κ^2 $[10^{-6}\,\mathrm{m^2}]$	B	In band?
1	Ván et al. [8]	Capacitor 3.9 mm	2.13 R	0.954 R	see note	2.23 R	no
2	Ván et al. [8]	Limestone 1.7 mm	2.950 R	0.991 R	see note	2.17 R	no
3	Ván et al. [8]	Foam 5.1 mm	2.373 R	0.402 R	see note	3.04 R	no
4	Ván et al. [8]	Leucocratic 1.75 mm	4.77 R	1.32 R	see note	1.77 R	**yes**
5	earlier eval.	Basalt 1.86 mm	0.55 R	0.738 R	0.509 R	1.243 R / 1.254 C	**yes**
6	earlier eval.	Basalt 2.75 mm	0.604 R	0.955 R	0.670 R	1.171 R / 1.162 C	**yes**
7	earlier eval.	Basalt 3.84 mm	0.68 R	0.664 R	0.480 R	1.060 R / 1.063 C	**yes**
8	Fehér & Kovács [27]	Basalt 1.86 mm	0.61 R	0.211 R	0.168 R	1.295 R / 1.305 C	**yes**
9	Fehér & Kovács [27]	Basalt 2.75 mm	0.61 R	0.344 R	0.268 R	1.272 R / 1.277 C	**yes**
10	Fehér & Kovács [27]	Basalt 3.84 mm	0.68 R	1.000 R	0.650 R	0.940 R / 0.956 C	**yes**
11	Fehér & Kovács [27]	Foam 5.2 mm	3.01 R	0.304 R	2.203 R	2.395 R / 2.408 C	no
12	Both et al. [7]	Capacitor 3.9 mm	1.958 R	0.51 R	1.53 R	1.532 C	**yes**
	reported standard errors: ± 0.004, ± 0.01, ± 0.03 respectively; $R^2 = 0.9996$						

Aggregate: 8 of 12 inside the band (44) (all six basalt evaluations, the leucocratic rock at $B = 1.77$ and the Both et al. capacitor at $B = 1.53$); 4 of 12 outside. **Uncertainties:** eleven of the twelve report no parameter uncertainty. Case 12 [7] is the exception and reports standard errors from nonlinear regression (0.2 % on α, 2.0 % on τ_q and on κ^2, $R^2 = 0.9996$); those errors are discussed in section 9.3. **Note (cases 1–4):** the authors tabulate a length scale rather than κ^2, and the reported dimensionless parameter could not be reproduced from it (limestone: 0.22 C against 2.17 R). The authors' reported values are used and the inconsistency is flagged, not reconciled. **Cases 5–7** were verified against Table 1 of [27], which tabulates the earlier values alongside the refined ones; the corresponding B values are given in its Table 2.

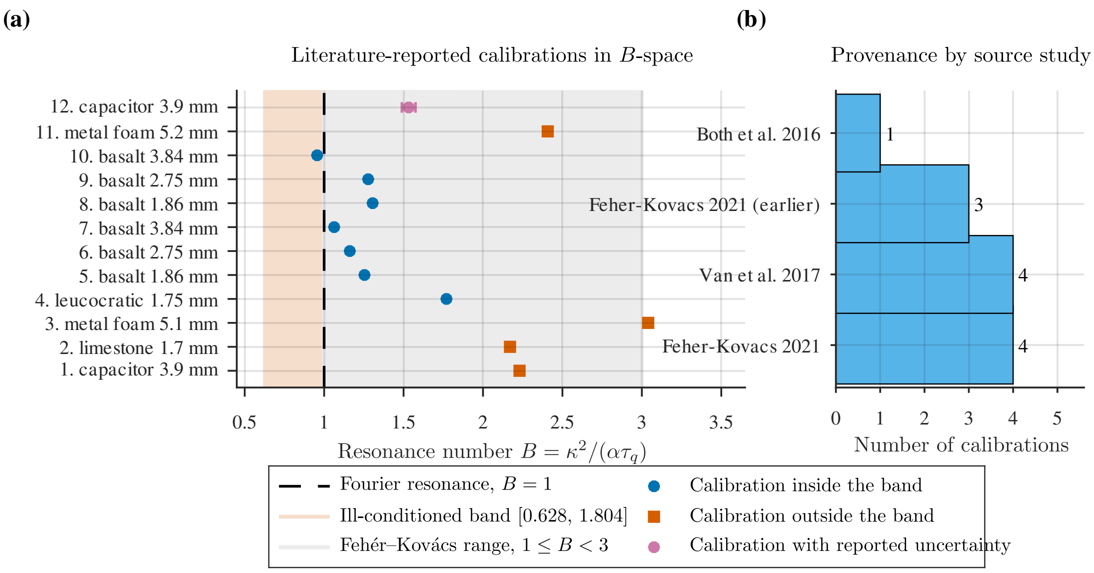

Figure 11: Literature-reported Guyer–Krumhansl calibrations placed in B-space. (a) The twelve audited calibrations against the verified ill-conditioned band $B \in [0.628, 1.804]$, within which s.e.(τ_q) exceeds 20%, and against the range $1 \le B < 3$ adopted by Fehér and Kovács. Calibrations falling inside the band are drawn as blue circles and those outside as vermilion squares; eight of the twelve lie inside. Only one calibration reports parameter uncertainties and only that case carries an error bar. The remaining eleven source studies do not report uncertainties and none has been synthesised for them. (b) Provenance of the twelve calibrations across the four source studies.

rock at $B = 1.77$, which falls just inside the upper edge, and the capacitor evaluation of Both et al. at $B = 1.53$. The remaining four, comprising the two metal foams, the limestone and the capacitor as evaluated by Ván et al., lie outside it at $B = 2.17$ to 3.04. The ill-conditioned band is therefore not a hypothetical construct: it contains a majority of the published calibrations of this model, so the identifiability structure bears directly on values already in circulation as material properties.

One feature of the audit requires explanation. Ván et al. [8] use a different nondimensionalisation from the one adopted here, scaling time by the pulse length rather than by L^2/α, and report a dimensionless group $\hat{b} = \hat{\ell}^2/(\hat{\tau}\hat{\alpha})$ which is algebraically identical to B. Combining their dimensionless table with their dimensional table does not reproduce their $\hat{b}$: for limestone that route gives 0.22 against a reported 2.17. Evaluated consistently within their dimensionless table alone, however, $\hat{\ell}^2/(\hat{\tau}\hat{\alpha})$ returns 1.99, 3.03 and 1.78 against reported values of 2.17, 3.04 and 1.77 for limestone, metal foam and leucocratic rock, agreeing to 0.01 for the latter two. The residual mismatch is traceable to the tabulated $\hat{\alpha}$ for limestone and leucocratic rock, which differ from $\alpha_{\mathrm{GK}} t_p/L^2$ computed from their own dimensional table by factors of 9.1 and 10.0 respectively, while the capacitor and metal foam entries agree to four significant figures. This is consistent with a units slip confined to two rows of the published table. The authors' reported B values are used here unchanged, and the two affected rows are flagged rather than silently recomputed.

Eleven of the twelve calibrations report no parameter uncertainty, correlation or confidence interval. The parameters are presented as point estimates of material properties, and the present analysis indicates that for eight of the twelve the relaxation time component of that estimate is weakly constrained by the measurement used to obtain it, under the observation design tested here. The single exception is treated separately in section 9.3.

9.3. *Calibration with reported uncertainties*

Both et al. [7] are the exception in the audited set. For the layered capacitor they report standard errors from a nonlinear regression on 2250 data points: $\alpha = [1.958 \pm 0.004] \times 10^{-6}\,\mathrm{m^2\,s^{-1}}$, $\tau_q = [0.51 \pm 0.01]\,\mathrm{s}$ and $\kappa^2 = [1.53 \pm 0.03] \times 10^{-6}\,\mathrm{m}^2$, with $R^2 = 0.9996$. These correspond to relative standard errors of 0.2 % on α and 2.0 % on both τ_q and κ^2, and place the specimen at $B = 1.53$, inside the band (44).

At first sight this contradicts the present analysis, which gives 27 % on τ_q at $B = 1.53$. It does not, because the two refer to different experiments. The band (44) is stated for the configuration of section 7.1, namely 80 observations at 2 % noise, whereas the cited regression uses 2250 points at substantially lower noise. Repeating the Fisher calculation at $B = 1.532$ with 2250 observations and 1 % noise returns a standard error of 2.55 % on τ_q, against the 2.0 % reported. The agreement of these two independently obtained figures is a check on the present framework against a real published dataset, and it illustrates concretely why the band is described throughout as configuration dependent rather than universal.

Two qualifications are needed. The reported standard errors are asymptotic regression errors computed, as the authors state, assuming exact L and t_p; section 7 shows that such errors track Monte Carlo spread well away from resonance but understate it close to it. And the specimen sits at $B = 1.53$, near the upper edge of the band rather than at its centre; the same measurement precision at $B \approx 1$ would not produce a comparable result, since the degeneracy there is exact.

9.4. *Same-specimen discrepancy*

An independent consistency check is available, because three basalt specimens were evaluated twice by different procedures. The two evaluations of the identical specimens return relaxation times differing by a factor of up to 3.50 (Table 7). The refined evaluation reports coefficients of determination above 0.99 for all three specimens [27]; a directly comparable R^2 for the earlier evaluation is not tabulated in the cited source and is not independently verified here.

Table 7: Two independent evaluations of the same three basalt specimens. The refined evaluation reports $R^2 > 0.99$ for the Guyer–Krumhansl fit for all three specimens; a comparable value is not tabulated for the earlier evaluation. All six lie at $B = 0.94$ to 1.31.

Specimen	τ_q earlier [s]	τ_q refined [s]	**Ratio**	R^2 (refined)
Basalt 1.86 mm	0.738	0.211	3.50×	0.9975
Basalt 2.75 mm	0.955	0.344	2.78×	0.9964
Basalt 3.84 mm	0.664	1.000	1.51×	0.9986

Two evaluations of the same data, each fitting well, returning relaxation times that differ by up to a factor of three is precisely the signature of a flat direction in the objective function. The magnitude is consistent with the present analysis: all six evaluations sit at $B = 0.94$ to 1.31, where the Monte Carlo of section 7 gives central 90% ranges for τ_q spanning factors of several to several tens, comfortably encompassing the observed spread.

The causal claim must nonetheless be limited. The observed parameter spread is consistent with and quantitatively accounted for by the identifiability structure near $B \approx 1$, while differences in pulse modelling, smoothing, and fitting procedures may also contribute. The two evaluations differ in more than one respect, and the present data do not permit the contributions to be separated. It is not claimed that the degeneracy caused the discrepancy.

10. Implications for Measurement Design

The analysis converts into a design criterion. Inverting the Fisher standard error of section 7 gives, for each target precision on τ_q, the region of B in which that target is attainable:

$$\text{s.e.}(\tau_q) \leq 10\% : \quad B \leq 0.441 \quad \text{or} \quad B \geq 3.464, \tag{47}$$
$$\text{s.e.}(\tau_q) \leq 20\% : \quad B \leq 0.628 \quad \text{or} \quad B \geq 1.804, \tag{48}$$
$$\text{s.e.}(\tau_q) \leq 30\% : \quad B \leq 0.723 \quad \text{or} \quad B \geq 1.468, \tag{49}$$
$$\text{s.e.}(\tau_q) \leq 50\% : \quad B \leq 0.817 \quad \text{or} \quad B \geq 1.252. \tag{50}$$

These boundaries hold for the configuration of section 7.1 and are not universal constants. They should be recomputed for any materially different observation design. The qualitative structure, an excluded interval centred on $B = 1$ whose width grows as the precision target tightens, is universal, because it follows from the structural result of section 3.

10.1. *Adjustable and fixed design variables*

B is composed of material properties and is not a free instrument setting. The available levers are correspondingly limited, and one commonly assumed lever does not exist.

Material selection.. The most effective lever is the choice of specimen. The audited calibrations of section 9 indicate that metal foams, at $B \approx 2.4$ to 3.0, and the limestone and capacitor samples at $B \approx 2.2$, are far better conditioned for relaxation time estimation than basalt at $B \approx 0.94$ to 1.31 or the leucocratic rock at $B = 1.77$. A preliminary estimate of B should therefore precede any campaign whose objective is to report τ_q.

Operating temperature.. The three coefficients α, τ_q and κ^2 are all temperature dependent, so B shifts with operating temperature. This provides a genuine, if material specific, means of moving a specimen away from resonance.

Specimen thickness does not move B.. This point is emphasised because the opposite is easy to assume. By (17), B is invariant under the scaling (11): the thickness cancels identically, and within the present one dimensional model all thicknesses of a given material share one value of B. This is confirmed by the audited data, where three basalt specimens of 1.86, 2.75 and 3.84 mm yield the same B to three significant figures from a single parameter set. Varying the thickness therefore cannot be used to escape the ill-conditioned band.

Multiple thicknesses as independent records.. What multiple thicknesses do provide is additional independent observations of the same material, which reduces the variance of a joint inversion in the ordinary way. This is a replication benefit, not a change in B, and it does not remove the structural degeneracy: at exactly $B = 1$ every record, at every thickness, is a Fourier record, so no number of them separates τ_q from κ^2.

Sampling and noise.. Denser sampling during the rise, where the sensitivity to τ_q is concentrated, and lower noise both reduce the standard error roughly in proportion to the usual factors, and so shift the band boundaries modestly inward. Neither removes the singularity, which is independent of noise level and sampling density by definition 1.

10.2. Recommended reporting practice

The single most useful change to reporting practice is inexpensive: report B alongside every fitted (τ_q, κ^2) pair. It is computed directly from the fitted parameters by (16), requires no additional measurement, and tells a reader immediately whether the reported relaxation time is supported by the data or is a point on a flat valley. Where B falls inside the band, the parameters should be reported in the (α, B) form of (30), with an interval rather than a point estimate for τ_q. Table 8 summarises the admissible B for each target precision together with the material classes of the audited set that satisfy it.

Table 8: Measurement design criterion. Admissible B for a target standard error on τ_q, for the configuration of section 7.1. These are not universal constants and should be recomputed for a different design. Thickness is not a lever: B is invariant under (17).

Target s.e.(τ_q)	**Admissible** B	**Example material class**
$\leq 10\%$	$B \leq 0.441$ or $B \geq 3.464$	none of the audited set
$\leq 20\%$	$B \leq 0.628$ or $B \geq 1.804$	metal foams, capacitor, limestone
$\leq 30\%$	$B \leq 0.723$ or $B \geq 1.468$	the above, plus leucocratic rock
$\leq 50\%$	$B \leq 0.817$ or $B \geq 1.252$	the above, plus basalt at 1.86 and 2.75 mm

11. Discussion

The principal finding is that a coincidence between two constitutive models, already known at the level of their solutions, has a consequence at the level of the inverse problem that does not follow from it automatically and that changes how published Guyer–Krumhansl coefficients should be read. The following subsections separate what was already established from what the present analysis adds, relate the structural and practical statements, and set out the consequences for calibration practice.

11.1. Established cancellation and its inverse-problem consequence

The Fourier resonance itself is established [26, 46, 47, 27]; no priority is claimed for it here. What this paper adds is the inverse-problem consequence: that two models coincide on a surface in parameter space is a statement about solutions, whereas the coincidence rendering a parameter pair unrecoverable from any temperature observation is a statement about the parameter-to-observable map, and the second does not follow from the first without the sensitivity analysis of section 3. The specific additions are the exact anti-parallel sensitivity relation (32) with its null direction $(1, \alpha)$, the consequent rank deficiency and Fisher singularity, the quantified width of the surrounding ill-conditioned region, the identification of (α, B) as the combination the data determine, and the resulting implications for calibration practice, model selection and measurement design.

11.2. Structural and practical identifiability

The two notions behave differently and the distinction carries the argument. The structural result is exact, holds at a single point, and is independent of noise, sampling and estimator; the practical result is approximate, holds over a finite region, and depends on all three. Neither alone would be sufficient: a point singularity in a three-parameter space would be a measure zero curiosity, while large standard errors without the exact result could always be attributed to an inadequate optimiser. It is the combination – an exact degeneracy at $B = 1$ broadening continuously into the band (44) – that makes the finding relevant to real calibrations, eight of the twelve audited here lying inside that band.

11.3. Reparameterisation rather than abandonment

The results are not an argument against the Guyer–Krumhansl model: it reproduces the measurements well, and the diffusivity and resonance number are recovered to about 1% and 10% even at the singularity itself. What fails is a specific act of interpretation – quoting τ_q and κ^2 as independent material constants when the observation constrains only a combination of them. The remedy is to report the combination the data determine, which is why section 2.9 introduces (30) as a device of the analysis rather than as a discovery.

11.4. Model-selection evidence

The Bayesian information criterion reaches the same conclusion by a route using neither derivatives nor a Fisher matrix: selecting the Fourier model at $B = 1$ from Guyer–Krumhansl-generated data is the rank deficiency expressed statistically, since the criterion cannot distinguish the additional parameters from the Fourier limit under the resonance condition. For experimentalists this is a caution rather than a diagnostic: a model comparison performed near resonance favours the simpler model regardless of the underlying physics, so a Fourier preference in that regime is evidence about the design, not the material.

11.5. Implications for published calibrations

The appropriate inference from the audited calibrations is about precision rather than correctness. The reported basalt relaxation times are not shown to be wrong; they are shown to be weakly constrained by the measurements from which they were obtained, whereas the reported diffusivities are well constrained throughout. This is consistent with, though not established as caused solely by, the identified degeneracy, since the two basalt evaluations also differ in pulse model, smoothing and fitting procedure.

The single calibration reporting uncertainties provides a check in the opposite direction: Both et al. [7] quote 2.0 % on τ_q at $B = 1.53$, and the present framework, evaluated at their sampling density and noise level, returns 2.55 %. Two independently obtained estimates agreeing at this level support the Fisher construction used throughout and show that a specimen inside the band can still yield a precise relaxation time given a sufficiently informative measurement – the band identifies where precision is expensive, not where it is unattainable.

11.6. Conditions under which the pair is separable

Separation requires the material to sit away from resonance, by a distance set by the target precision: for the design of section 7.1, a 20% standard error on τ_q requires $B \leq 0.628$ or $B \geq 1.804$, and a 10% error requires $B \leq 0.441$ or $B \geq 3.464$ – satisfied by no specimen in the audited set. Two levers act on this condition: specimen selection, since B is a material property (the metal foams of the audited set, at $B \approx 2.4$–3.0, are the best-conditioned available), and measurement quality, which shifts the band boundaries without moving B (the capacitor calibration of Both et al. [7] attains 2.0% at $B = 1.53$, inside the reference 20% band, using 2250 low-noise observations). Specimen thickness is not a lever, since B is invariant under (17). The practical criterion is therefore to estimate B from a preliminary fit before committing to a relaxation-time measurement.

11.7. Regime of validity

The analysis concerns the Guyer–Krumhansl equation as an effective continuum description of heterogeneous media at room temperature, with coefficients constrained by the second law rather than derived from a carrier model [15, 8] – the regime of the audited calibrations. Conclusions are not extended to the phonon-hydrodynamic regime in dielectric crystals at low temperature, where the coefficients have a kinetic interpretation obtainable independently of a flash-experiment fit and the inverse problem examined here does not arise in the same form.

12. Scope and Limitations

One-dimensional formulation.. The analysis is carried out for the one dimensional slab of section 2. The cancellation of proposition 1 is a property of the propagation factor $m^2(s)$, which arises in the same form in geometries that separate, so the structural result is expected to persist there; a treatment of genuinely multi dimensional geometries, including lateral losses and finite beam profiles, has not been carried out and is not claimed.

Inverse experiments are synthetic.. The identifiability results are established on synthetic data. This is necessary rather than incidental: demonstrating that two parameter sets are indistinguishable requires knowing the parameter values that generated the data, which no experiment provides. The synthetic study is anchored to reality by using physical basalt parameters, by auditing published calibrations in section 9, and by the same specimen discrepancy of table 7, which is real data behaving as the analysis predicts. It remains a limitation that the inverse conclusions have not been tested against a controlled experiment with independently known parameters.

The band is configuration dependent.. The numerical boundaries (43)–(46) and the design criterion (47)–(50) hold for the configuration of section 7.1. They are not physical constants and would move under a different sampling scheme, noise level, or pulse length. Only the singularity at $B = 1$ is universal.

Forward validation is not inverse validation.. section 5 validates the forward operator against published figures. It does not validate the inverse problem, and is not offered as doing so. The inverse results rest on the analytical proof of section 3 together with the numerical certification of section 6 and the independent statistical route of section 8.

Dependence on the pulse and boundary model.. The structural result is independent of the excitation, since $\hat{Q}_0(s)$ cancels from the sensitivity ratio (34). The width of the practically ill-conditioned band is not: it depends on the pulse shape and duration through the information content of the record. Boundary cooling, neglected here as in the anchor, would modify the observable and hence the band boundaries, though not the degeneracy.

Unverified bibliographic and calibration items.. For the four samples of Ván et al. the reported dimensionless parameter could not be reproduced from the tabulated length scale, as recorded in section 9.2; the discrepancy is flagged and not reconciled. The three earlier basalt evaluations were verified against Table 1 of [27], which reproduces them; they were not traced further back to the measurement report in which they originally appeared. Sections 4 and 5 of the version of record of the anchor are behind a paywall, and the equations used here were transcribed from the open preprint; the version of record is Volume 127, Part A, and its equation numbering differs from the preprint, which is why no equation numbers of the anchor are cited in section 2.10.

Raw experimental data unavailable.. The published heat pulse time series underlying the audited calibrations are not available, so the re-analysis of section 9 necessarily works from tabulated fitted values rather than from the measurements themselves. A direct re-inversion of the original records, with uncertainty quantification, would be a stronger test and is not possible with the published material.

Literature screening was not performed on an authenticated citation index.. Access to Scopus and Web of Science was not available. The screening reported in section 1 used open bibliographic sources with validated positive controls, covered title and abstract records rather than full text, and achieved abstract coverage of about 60% of the retrieved corpus. Statements about the absence of prior work are therefore statements about the screened corpus and not absolute priority claims. An authenticated index search remains outstanding.

Local optima in the profile computation.. The profile likelihood is computed by re-optimising the remaining parameters at each node of a fixed τ_q grid with a two-start local method. At $B = 0.90$ this procedure does not return a unimodal profile, as recorded in section 7.3, so no confidence interval is quoted at that condition. A global search at each node would resolve it. The conclusions rest on the four conditions at which the profile is unimodal, on the Fisher analysis and on the Monte Carlo study, none of which requires re-optimisation at fixed τ_q.

A superseded numerical result.. As detailed in Appendix D, an earlier implementation produced a larger optimiser-initialisation effect than the certified solver reproduces; that earlier result is not used. Similarly, an earlier reported discrepancy between Fisher and Monte Carlo uncertainties was traced to finite-difference Jacobians and withdrawn; section 7 reports agreement between the two where the asymptotic approximation is valid.

13. Conclusions

The parameter-to-observable map of the one-dimensional Guyer–Krumhansl heat pulse problem has been analysed as an inverse problem, using a Laplace-domain convolution solver with analytic sensitivities, cross-checked against an independent finite-difference scheme and an arbitrary-precision reference, and externally validated against three published figures of an independently authored analytical solution at normalised errors of 0.384%, 0.382% and 0.483% with no parameter fitting.

Three findings follow. First, at the Fourier resonance $B = \kappa^2/(\alpha\tau_q) = 1$, a condition established in the existing literature and not claimed here, the response is independent of the relaxation time at every position and time and for any excitation; the sensitivities satisfy $\partial T/\partial\tau_q = -\alpha\,\partial T/\partial\kappa^2$ exactly, so the sensitivity matrix has rank one with null direction $(1, \alpha)$, the Fisher matrix is singular, and τ_q and κ^2 are structurally non-identifiable from any temperature record at any noise level or sampling density. Second, the degeneracy is not confined to that point: for the tested configuration the relaxation time carries a standard error above 20% throughout $B \in [0.628, 1.804]$ and above 50% throughout $B \in [0.817, 1.252]$; at $B = 1$ specifically, within this ill-conditioned region, the Bayesian information criterion selects the Fourier model over the Guyer–Krumhansl model even for Guyer–Krumhansl data, while away from resonance the Guyer–Krumhansl model is selected decisively at every tested condition, including $B = 1.28$ inside the 20% band. Third, the record nevertheless remains informative: the diffusivity is recovered to about 1% and the resonance number to between 4% and 10% across the whole range including exactly at resonance, so what fails is the resolution of a determined combination into two coefficients rather than the determination of the transport behaviour.

The contribution is therefore a statement about estimation rather than constitutive modelling: a coincidence already known at the level of solutions is shown to impose an exact rank deficiency on the inverse problem, with the neighbourhood in which that deficiency dominates finite-noise estimation quantified. The practical consequence is concrete: of twelve audited published calibrations, eight lie inside the band and eleven report no parameter uncertainty, and two independent evaluations of the same three basalt specimens return relaxation times differing by factors of 1.51 to 3.50 at $R^2 > 0.99$, consistent with, though not attributable solely to, the identified structure. Guyer–Krumhansl coefficients are best reported in the (α, B) parameterisation, and campaigns targeting the relaxation time should select specimens away from resonance, since specimen thickness leaves B invariant.

The analysis is confined to the one-dimensional slab and to synthetic inverse experiments; the band boundaries are properties of the stated observation design and only the singularity at their centre is universal. No experimental validation of the inverse conclusions is claimed. The most useful extension is a controlled heat pulse experiment on a specimen of independently determined B, which would test the predicted precision directly and, for a specimen placed away from resonance, establish whether the relaxation and nonlocal coefficients can be separated in practice at the precision the present analysis predicts.

CRediT authorship contribution statement

V. Gupta: Conceptualization, Methodology, Software, Formal analysis, Validation, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The numerical implementation, the MATLAB/Octave figure-generation scripts, and the verified numerical results underlying every figure and table in this paper are available from the corresponding author upon reasonable request. [REPOSITORY DOI OR URL – TO BE SUPPLIED BY AUTHOR upon deposit, e.g. Zenodo or an institutional repository, prior to submission.]

Appendix A. Numerical Algorithm

Algorithm 1 gives the complete procedure summarised in section 4.6.

Algorithm 1 Forward evaluation, sensitivity assembly and identifiability diagnostics for the Guyer–Krumhansl rear-face problem.

Require: diffusivity α, relaxation time τ_q, nonlocal coefficient κ^2, slab thickness L, pulse length t_p, sample times $\{t_i\}_{i=1}^{n}$, noise level η

Ensure: rear-face history $\hat{T}(1, t_i)$, Fisher matrix F, standard errors, profile likelihood, model-selection scores

Stage 1: dimensionless reduction

$B \leftarrow \kappa^2/(\alpha\tau_q), \quad \tau_\Delta \leftarrow L^2/\alpha, \quad \hat{t}_i \leftarrow t_i/\tau_\Delta$ ▷ section 2.4

Stage 2: forward solution (Solver A)

Form the propagation function $m^2(s) = s\,(1+\tau_q s)/(\alpha+\kappa^2 s)$ ▷ eq. (21)

Factorise $\hat{T}(1,s) = \hat{K}(s)\,\hat{Q}_0(s)$ and retain *only* $\hat{K}$ for numerical inversion ▷ $\hat{Q}_0$ carries removable poles

Invert $\hat{K}$ on a fixed Talbot contour with $M = 41$ nodes ▷ M odd; see section 6.1

Convolve the inverted kernel with the analytic pulse q_0 by Gauss–Legendre quadrature, $n_q = 32$

return $\hat{T}(1, \hat{t}_i)$

Stage 3: independent cross-checks

Solve the same problem with Solver B, a staggered-grid finite-difference scheme at $N_x = 400$, and with Solver C at 50-digit precision

assert pairwise agreement within the criteria of table C.1 ▷ numerical cross-checking, not external validation

Stage 4: analytic sensitivities

for $\theta \in \{\alpha,\ \tau_q,\ \kappa^2\}$ **do**

  Differentiate $m^2(s)$ in closed form and invert $\partial\hat{K}/\partial\theta$ on the same contour ▷ no finite differencing

end for

Assemble the Jacobian $J_{i\theta} = \partial\hat{T}(1,\hat{t}_i)/\partial\theta$

Stage 5: identifiability diagnostics

$F \leftarrow \sigma^{-2} J^\top J$; evaluate $\mathrm{cond}(\tilde{F})$, $\rho_{\tau_q\kappa^2}$ and $\mathrm{s.e.}(\theta) = \sqrt{[F^{-1}]_{\theta\theta}}$

if $\mathrm{cond}(\tilde{F}) > 10^{12}$ **then**

  report the pair as numerically non-identifiable at this B

end if

Profile the likelihood in τ_q on a fixed grid, re-optimising the remaining parameters at each node, and intersect with the χ^2_1 threshold 3.841

Repeat the estimation over 200 Monte Carlo noise realisations and compare the sampling spread with the Fisher prediction

Stage 6: model comparison

for each candidate $\mathcal{M} \in \{\text{Fourier, MCV, Nyíri, GK}\}$ **do**

  Fit $\mathcal{M}$ to the same synthetic record by Levenberg–Marquardt with three restarts

  $\mathrm{BIC}_\mathcal{M} \leftarrow n\ln(\mathrm{SSE}/n) + k\ln n$ ▷ k = number of free parameters

end for

$\Delta\mathrm{BIC} \leftarrow \mathrm{BIC} - \min_\mathcal{M}\mathrm{BIC}$ and select $\arg\min$

Appendix B. Notation Correspondence with the Validation Anchor

Table B.1 lists the correspondence between the notation used here and that of the validation anchor [26], referenced from sections 2.10 and 5.

Table B.1: Correspondence between the present notation and that of the validation anchor [26]. Symbols are matched by the quantity they denote, not by appearance.

Present	**Anchor**	**Quantity**	**Conversion**
α	α	thermal diffusivity $\lambda/(\rho c)$	identical
τ_q	τ_q	relaxation time of the heat flux	identical
κ^2	κ^2	coefficient of $\partial_t \partial_{xx} T$, dimension m^2	identical
τ_Δ	$\hat{\tau}_\Delta$	dimensionless pulse length	$\alpha t_p/L^2$
$\hat{\tau}_q$	$\hat{\tau}_q$	scaled relaxation time	$\alpha \tau_q/L^2$
$\hat{\kappa}^2$	$\hat{\kappa}^2$	scaled nonlocal coefficient	κ^2/L^2; the anchor also writes $\hat{\kappa} = \kappa/L$
B	$\kappa^2/(\alpha\tau_q)$	Fourier-resonance number; the anchor states the resonance as $\kappa^2/\tau_q = \alpha$	$B = \kappa^2/(\alpha\tau_q)$
$m(s)$	not used	Laplace propagation factor; the anchor uses an eigenfunction expansion	introduced here

Appendix C. Supplementary Numerical Certification

Table C.1 itemises the complete numerical error budget underlying the certification summarised in section 6 and illustrated in figs. 5 and 6.

Table C.1: Numerical certification and error budget. Each entry states the scope over which it was established. Agreement between the present solvers is numerical cross-checking, not external validation.

Item	Result	Scope
Talbot node parity	failure at $t^* = M\tau_\Delta/10$ for even M	split formulation; error 10^9–10^{11} at t^*, 10^{-4} adjacent; absent for odd M
Convolution formulation	no pole encountered	kernel $\hat{K}$ is pole free; adopted throughout
Talbot node count	plateau $M \in [25, 81]$; $M = 41$ used	degrades beyond $M \approx 91$ by round off
Convolution quadrature	$\leq 10^{-13}$	32 versus 64 Gauss–Legendre nodes
Solver A versus Solver C	10^{-11} to 10^{-13}	probe times away from the steep rise
Solver A versus Solver B	$\approx 10^{-7}$	full record; limited by finite differences
Finite difference, space	observed order $p = 2.00$; 2.6×10^{-6}	$N_x = 400$ against Richardson reference
Finite difference, time	$\approx 10^{-9}$	relative tolerance 10^{-10}
Arithmetic precision	$\leq 8.9 \times 10^{-16}$	15, 30 and 50 digits compared
Fourier-resonance limit	2.6×10^{-6}	$\kappa^2 = \alpha\tau_q$ against analytical Fourier solution
Non-negativity, steady state	$T \geq 0$; $T(\infty) = 1.00000000$	all cases tested
Total numerical error	$\approx 10^{-6}$	versus digitisation floor 6×10^{-3}

Appendix D. Optimiser-Initialisation Sensitivity

This appendix gives the complete optimiser-initialisation sensitivity study summarised in section 7.6. Starting points were drawn from log-normal perturbations of the true parameters at $B = 1.277$, and the spread of the recovered values was recorded separately from the spread induced by measurement noise.

Table D.2: Optimiser-initialisation sensitivity at $B = 1.277$. The central 90% range of recovered τ_q and B is reported as a multiple of the true value, for increasing perturbation of the optimiser starting point.

Perturbation	τ_q **range (central 90%)**	B **range**
0.5 log units	essentially unchanged	0.91 to 1.28
1.0 log units	factor $4.7 \to 9.3$	0.91 to 1.28

With the certified solver the effect is modest and, in these tests, smaller than the effect of measurement noise itself; a small number of realisations converge to a distant local minimum rather than following the trend in the table. Parameter estimates can nonetheless exhibit substantial dependence on optimiser initialisation in poorly conditioned regimes, motivating multi-start or global search strategies. An implementation of the Laplace inversion based on the split pulse formulation, whose failure at $t^* = M\tau_\Delta/10$ is characterised in section 6.1, exhibits a substantially larger effect, a spread of roughly a factor of twenty-three in recovered τ_q, because for the settings then in use the failure time fell inside the observation window; the result in the table, obtained with the certified convolution solver, supersedes this and is the value used throughout the paper.

References

[1] S. Sobolev, Non-Fourier heat conduction: Discrete vs continuum approaches, Mechanics Research Communications 149 (2025) 104512. doi:10.1016/j.mechrescom.2025.104512.

[2] A. J. Moghadam, Characterizing the thermal penetration depth of non-Fourier heat conduction in an infinite medium, International Communications in Heat and Mass Transfer 161 (2025) 108522. doi:10.1016/j.icheatmasstransfer.2024.108522.

[3] V. Gupta, M. S. Barak, S. Das, Impact of memory-dependent heat transfer on Rayleigh waves propagation in nonlocal piezo-thermo-elastic medium with voids, International Journal of Numerical Methods for Heat & Fluid Flow 34 (4) (2024) 1902–1926. doi:10.1108/HFF-10-2023-0615.

[4] V. Gupta, M. S. Barak, Photo-thermo-piezo-elastic waves in semiconductor medium subject to distinct two temperature models with higher order memory dependencies, International Journal of Numerical Methods for Heat & Fluid Flow 34 (1) (2024) 84–108. doi:10.1108/HFF-07-2023-0380.

[5] V. Gupta, R. Kumar, R. Kumar, M. S. Barak, Energy analysis at the interface of piezo/thermoelastic half spaces, International Journal of Numerical Methods for Heat & Fluid Flow 33 (2023) 2250–2277. doi:10.1108/HFF-11-2022-0654.

[6] V. Gupta, R. Kumar, M. Kumar, V. Pathania, M. S. Barak, Reflection/transmission of plane waves at the interface of an ideal fluid and nonlocal piezothermoelastic medium, International Journal of Numerical Methods for Heat & Fluid Flow 33 (2) (2023) 912–937. doi:10.1108/HFF-04-2022-0259.

[7] S. Both, B. Czél, T. Fülöp, G. Gróf, Á. Gyenis, R. Kovács, P. Ván, J. Verhás, Deviation from the Fourier law in room-temperature heat pulse experiments, Journal of Non-Equilibrium Thermodynamics 41 (1) (2016) 41–48. doi:10.1515/jnet-2015-0035.

[8] P. Ván, A. Berezovski, T. Fülöp, G. Gróf, R. Kovács, Á. Lovas, J. Verhás, Guyer–Krumhansl-type heat conduction at room temperature, EPL 118 (5) (2017) 50005. doi:10.1209/0295-5075/118/50005.

[9] R. A. Guyer, J. A. Krumhansl, Solution of the linearized phonon Boltzmann equation, Physical Review 148 (2) (1966) 766–778. doi:10.1103/PhysRev.148.766.

[10] R. A. Guyer, J. A. Krumhansl, Thermal conductivity, second sound and phonon hydrodynamic phenomena in nonmetallic crystals, Physical Review 148 (2) (1966) 778–788. doi:10.1103/PhysRev.148.778.

[11] M. Xu, The modified Guyer-Krumhansl equations derived from the linear Boltzmann transport equation, ASME Journal of Heat and Mass Transfer 146 (2024) 102501. doi:10.1115/1.4065818.

[12] M. Di Domenico, D. Jou, Nonlinear phonon hydrodynamics: A guyerkrumhansl equation for the conjugate of the heat flux, Zeitschrift fur angewandte Mathematik und Physik 76 (2025) 235. doi:10.1007/s00033-025-02630-7.

[13] L. Restuccia, D. Jou, Non-local vectorial internal variables and generalized Guyer-Krumhansl evolution equations for the heat flux, Entropy 25 (2023) 1259. doi:10.3390/e25091259.

[14] R. Kovacs, Interplay of time scales and material nonlinearities in non-Fourier heat conduction, Continuum Mechanics and Thermodynamics 38 (2026) 46. doi:10.1007/s00161-026-01477-1.

[15] P. Ván, Weakly nonlocal irreversible thermodynamics: The Guyer–Krumhansl and the Cahn–Hilliard equations, Physics Letters A 290 (1–2) (2001) 88–92. doi:10.1016/S0375-9601(01)00657-0.

[16] M. Szucs, M. Pavelka, R. Kovacs, T. Fulop, P. Van, M. Grmela, A case study of non-Fourier heat conduction using internal variables and generic, Journal of Non-Equilibrium Thermodynamics 47 (2021) 31–60. doi:10.1515/jnet-2021-0022.

[17] S. A. Rukolaine, Hyperbolicity of the ballistic-conductive model of heat conduction: the reverse side of the coin, Zeitschrift fur angewandte Mathematik und Physik 75 (2024) 50. doi:10.1007/s00033-023-02176-6.

[18] K. Zhukovsky, Exact solution of guyerkrumhansl type heat equation by operational method, International Journal of Heat and Mass Transfer 96 (2016) 132–144. doi:10.1016/j.ijheatmasstransfer.2016.01.005.

[19] K. Zhukovsky, Violation of the maximum principle and negative solutions for pulse propagation in guyerkrumhansl model, International Journal of Heat and Mass Transfer 98 (2016) 523–529. doi:10.1016/j.ijheatmasstransfer.2016.03.021.

[20] D. Maillet, A review of the models using the Cattaneo and Vernotte hyperbolic heat equation and their experimental validation, International Journal of Thermal Sciences 139 (2019) 424–432. doi:10.1016/j.ijthermalsci.2019.02.021.

[21] R. Kovacs, P. Rogolino, Numerical treatment of nonlinear Fourier and Maxwell-Cattaneo-Vernotte heat transport equations, International Journal of Heat and Mass Transfer 150 (2020) 119281. doi:10.1016/j.ijheatmasstransfer.2019.119281.

[22] A. van der Merwe, N. van Rensburg, R. Sieberhagen, Comparing the dual phase lag, Cattaneo-Vernotte and Fourier heat conduction models using modal analysis, Applied Mathematics and Computation 396 (2021) 125934. doi:10.1016/j.amc.2020.125934.

[23] B. Nyíri, On the entropy current, Journal of Non-Equilibrium Thermodynamics 16 (2) (1991) 179–186. doi:10.1515/jnet.1991.16.2.179.

[24] X. Zhang, X. Zhang, A high-order compact alternating direction implicit difference scheme of for solving non-Fourier dual-phase-lag equation with a moving heat source, International Communications in Heat and Mass Transfer 159 (2024) 108152. doi:10.1016/j.icheatmasstransfer.2024.108152.

[25] A. Fehér, N. Lukács, L. Somlai, T. Fodor, M. Szücs, T. Fülöp, P. Ván, R. Kovács, Size effects and beyond-Fourier heat conduction in room-temperature experiments, Journal of Non-Equilibrium Thermodynamics 46 (4) (2021) 403–411. doi:10.1515/jnet-2021-0033.

[26] R. Kovács, Analytic solution of Guyer–Krumhansl equation for laser flash experiments, International Journal of Heat and Mass Transfer 127 (Part A) (2018) 631–636. doi:10.1016/j.ijheatmasstransfer.2018.06.082.

[27] A. Fehér, R. Kovács, On the evaluation of non-Fourier effects in heat pulse experiments, International Journal of Engineering Science 169 (2021) 103577. doi:10.1016/j.ijengsci.2021.103577.

[28] S. Saurav, S. Mazumder, Anisotropic Fourier heat conduction and phonon Boltzmann transport equation based simulation of time domain thermo-reflectance experiments, International Journal of Heat and Mass Transfer 228 (2024) 125698. doi:10.1016/j.ijheatmasstransfer.2024.125698.

[29] J. Zheng, F. Li, H. Huang, T-phpinn: Physics-informed neural networks for solving 2d non-Fourier heat conduction equations, International Journal of Heat and Mass Transfer 235 (2024) 126216. doi:10.1016/j.ijheatmasstransfer.2024.126216.

[30] S. Li, R. E. Mohan, J. Wang, H.-S. Chu, S. Shin, Quasi-ballistic heat conduction of long-range surface phonon polaritons coupled to thermal reservoirs, ACS Nano 19 (2025) 21020–21029. doi:10.1021/acsnano.5c04594.

[31] L. Kremeyer, T. L. Britt, B. J. Siwick, S. C. Huberman, Ultrafast electron diffuse scattering as a tool for studying phonon transport: Phonon hydrodynamics and second sound oscillations, Structural Dynamics 11 (2024) 024101. doi:10.1063/4.0000224.

[32] H. Najafi, K. Woodbury, K. Dowding, Special issue: First james v. beck memorial symposium on inverse problems, parameter estimation, and heat conduction, ASME Journal of Heat and Mass Transfer 146 (2024) 090301. doi:10.1115/1.4066037.

[33] H. Long, Y. Zhang, Stochastic asymptotical regularization for nonlinear ill-posed problems, Inverse Problems 41 (2025) 065017. doi:10.1088/1361-6420/addffc.

[34] Y. Shang, H. Ban, D. Liu, Simultaneous identification of boundary heat flux and thermal conductivity in inverse heat conduction problems using physics-informed neural networks, Thermal Science and Engineering Progress 65 (2025) 103905. doi:10.1016/j.tsep.2025.103905.

[35] V. T. Borukhov, G. M. Zayats, The inverse problem of heat conduction in the case of non-uniqueness: A functional identification approach, Journal of Inverse and Ill-posed Problems (2024). doi:10.1515/jiip-2022-0056.

[36] A. Raue, C. Kreutz, T. Maiwald, J. Bachmann, M. Schilling, U. Klingmuller, J. Timmer, Structural and practical identifiability analysis of partially observed dynamical models by exploiting the profile likelihood, Bioinformatics 25 (2009) 1923–1929. doi:10.1093/bioinformatics/btp358.

[37] M. J. Simpson, O. J. Maclaren, Profile-wise analysis: A profile likelihood-based workflow for identifiability analysis, estimation, and prediction with mechanistic mathematical models, PLOS Computational Biology 19 (2023) e1011515. doi:10.1371/journal.pcbi.1011515.

[38] I. Borisov, A. Demin, E. Metelkin, Likelihoodprofiler.jl: Unified profile-likelihood workflows for identifiability and confidence intervals, Journal of Open Source Software 11 (2026) 9501. doi:10.21105/joss.09501.

[39] M. J. Simpson, R. E. Baker, Parameter identifiability, parameter estimation, and model prediction for differential equation models, SIAM Review 68 (2026) 153–171. doi:10.1137/24m1667968.

[40] M. Binns, A. Usai, C. Theodoropoulos, Identifiability methods for biological systems: Determining subsets of parameters through sensitivity analysis, penalty-based optimisation, profile likelihood and lasso model reduction, Computers & Chemical Engineering 186 (2024) 108683. doi:10.1016/j.compchemeng.2024.108683.

[41] Y. Liu, P. K. Maini, R. E. Baker, Optimal experiment design for practical parameter identifiability and model discrimination, Mathematical Biosciences 399 (2026) 109710. doi:10.1016/j.mbs.2026.109710.

[42] Y. Jung, I. Lee, Optimal design of experiments for optimization-based model calibration using Fisher information matrix, Reliability Engineering & System Safety 216 (2021) 107968. doi:10.1016/j.ress.2021.107968.

[43] E. A. Stromberg, J. Nyberg, A. C. Hooker, The effect of Fisher information matrix approximation methods in population optimal design calculations, Journal of Pharmacokinetics and Pharmacodynamics 43 (2016) 609–619. doi:10.1007/s10928-016-9499-4.

[44] T. B. Neilsen, M. K. Transtrum, D. F. Van Komen, D. P. Knobles, Optimal experimental design for machine learning using the Fisher information matrix, The Journal of the Acoustical Society of America 144 (2018) 1730–1730. doi:10.1121/1.5067675.

[45] C. Wang, Y. Heng, J. Luo, X. Wang, A fast Bayesian parallel solution framework for large-scale parameter estimation of 3d inverse heat transfer problems, International Communications in Heat and Mass Transfer 155 (2024) 107409. doi:10.1016/j.icheatmasstransfer.2024.107409.

[46] P. Ván, R. Kovács, T. Fülöp, Thermodynamic hierarchies of evolution equations, Proceedings of the Estonian Academy of Sciences 64 (3S) (2015) 389–395. doi:10.3176/proc.2015.3S.09.

[47] T. Fülöp, R. Kovács, Á. Lovas, Á. Rieth, T. Fodor, M. Szücs, P. Ván, G. Gróf, Emergence of non-Fourier hierarchies, Entropy 20 (11) (2018) 832. doi:10.3390/e20110832.

[48] R. Kovács, P. Ván, Generalized heat conduction in heat pulse experiments, International Journal of Heat and Mass Transfer 83 (2015) 613–620. doi:10.1016/j.ijheatmasstransfer.2014.12.045.

[49] Á. Rieth, R. Kovács, T. Fülöp, Implicit numerical schemes for generalized heat conduction equations, International Journal of Heat and Mass Transfer 126 (2018) 1177–1182. doi:10.1016/j.ijheatmasstransfer.2018.06.067.

[50] A. Talbot, The accurate numerical inversion of Laplace transforms, IMA Journal of Applied Mathematics 23 (1979) 97–120. doi:10.1093/imamat/23.1.97.

[51] B. Dingfelder, J. A. C. Weideman, An improved Talbot method for numerical Laplace transform inversion, Numerical Algorithms 68 (2014) 167–183. doi:10.1007/s11075-014-9895-z.

[52] H. Chhajer, R. Roy, Rationalised experiment design for parameter estimation with sensitivity clustering, Scientific Reports 14 (2024) 25864. doi:10.1038/s41598-024-75539-2.

[53] D. L. Weakliem, A critique of the Bayesian information criterion for model selection, Sociological Methods & Research 27 (1999) 359–397. doi:10.1177/0049124199027003002.

[54] K. C. Woo, L. C. Chow, Inverse heat conduction by direct inverse Laplace transform, Numerical Heat Transfer 4 (1981) 499–504. doi:10.1080/01495728108961807.

[55] V. V. Kryzhniy, Numerical inversion of the Laplace transform: analysis via regularized analytic continuation, Inverse Problems 22 (2006) 579–597. doi:10.1088/0266-5611/22/2/012.

[56] V. Zakian, Numerical inversion of Laplace transform, Electronics Letters 5 (1969) 120–121. doi:10.1049/el:19690090.

[57] S. A. Rukolaine, Unphysical effects of the dual-phase-lag model of heat conduction, International Journal of Heat and Mass Transfer 78 (2014) 58–63. doi:10.1016/j.ijheatmasstransfer.2014.06.066.